

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES
TOWARD STARS WITHIN 40 PC AT 114–873 GHzGLENN J. WHITE^{1,2} AND ROBIN DEY³

ABSTRACT

The published radio technosignature effort stops near 30 GHz; no search above it has reached a star within 40 pc. The ALMA archive reaches that band incidentally: observations of dust discs and planet formation record a spectrum of the star at every field centre. Re-reduced uniformly from the raw visibilities they search for a drifting, unresolved carrier at the stellar position in 402 fine-channel windows toward 60 stellar systems, with a coarse-channel bound on an unresolved spectral excess that discriminates no carrier. Nothing met the confirmation criterion. Of the 56 threshold crossings, four arise in one execution block that fails a quality criterion fixed in advance, and 18 of the rest coincide with a catalogued molecular transition, among them β Pictoris’s carbon monoxide, recovered blind. Every unattributed crossing was tested for recurrence in the star’s own frame, in each later covering window at every drift searched; none returned, including the two that lead their own control fields. That excludes a carrier holding its frequency inside the window searched and radiating through both observations, not a persistent transmitter of any kind; and the remainder is consistent with chance, to the accuracy with which control positions stand in for the star. Depth is a recovery fraction, not a threshold: injected carriers come back nine times in ten at 1.6×10^{15} W for the median system, over a union bandwidth about twice what recurrence could test. Molecular-line confusion and sparse repeat coverage, not thermal sensitivity, limit a search of this kind.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer’s correlator setup. This paper searches those spectra for a narrow-band transmitter at the star’s own position. Nothing of the kind survived.

Searches for *technosignatures* — detectable imprints of technology on the electromagnetic spectrum, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet the last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018), and essentially all of it lies below ~ 30 GHz. No published radio technosignature search above that frequency has reached a star within 40 pc, although Mason et al. (2025) have searched more distant stars. The band is attractive because a carrier arrives less distorted and a given aperture delivers more gain per watt supplied, and hard because archived correlator channels are far coarser than a dedicated narrowband backend and are crowded with molecular lines — a dense background of real astrophysical signals that a threshold search must separate from an artificial one (§2).

The signal sought is an *unresolved spectral carrier*: emission confined to a few channels, unresolved at the stellar position, and present at more than one epoch. Each qualifier is a restriction, and the last is the strictest: it is recurrence rather than brightness that separates an interesting event from a detection (Sheikh et al. 2021). The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body (§4.1). “Narrowband” here describes the assumed transmitter and never the resolving power of the archived channels, which are fine only by the standards of this archive. The targets were chosen by other programmes for other purposes, so every statistical statement that follows is about an archive-selected sample (§6.3).

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024), several of them riding back-ends built for other science (Morrison et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay & Tingay 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The classical threshold search used here lacks that capability, and instead uses a spa-

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tial test on the star against control positions in the same field, which a single dish cannot make. A narrowband search also reaches only one class of signal: broadband planetary-scale leakage (Garrett 2026) is outside it by construction.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA observes in Bands 3 to 10, spanning roughly 84 to 950 GHz; of those eight bands, one has been searched for technosignatures. For continuity with the centimetre literature we note one familiar power scale: the Arecibo planetary radar radiated 2.0×10^{13} W at 2.38 GHz (Sheikh et al. 2025). It carries no transmitter model into the millimetre band, where an aperture of that diameter is not realisable, so the illustrative benchmark of §6.3 is instead a dish of 12 m. Neither is a privileged scale; the limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search and the only comparable archival one. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\text{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies $25\times$ further away than the most distant star searched here for a carrier and $277\times$ further than the nearest of those: the two samples differ by 1.4–2.4 orders of magnitude in distance, and at a fixed received flux by the square of that, which is 3–5 orders in minimum detectable EIRP. An EIRP ratio between the two surveys would therefore be almost entirely the d^2 term. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min} \delta\nu$. It must be quoted at a matched channel width, because $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$. Mason et al. reach 3.8×10^{-23} – 5.8×10^{-23} W m $^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same channel width and the same threshold convention, the fine-channel windows searched here reach 5.2×10^{-24} W m $^{-2}$ at best, a factor 9 deeper, and 7.1×10^{-23} W m $^{-2}$ at the median, a factor 1.5 shallower. The two experiments are therefore comparable in depth and differ in what they cover: distant stars in four calibrator fields, against 60 stellar systems within 40 pc searched for a drifting carrier, over a frequency range reaching about nine times higher. They also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the dense forest of molecular lines raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so for any star in common the numbers to cite are this paper’s.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. The vertical axis is physically commensurable: for a carrier narrower than every programme’s channel, the minimum detectable EIRP is a total radiated power and not a spectral power density, so the same quantity is plotted for each programme. What differs is the convention, so both are drawn. Every published limit shown is a nominal 5σ threshold at which recovery is not stated, and this

survey’s filled symbols are the same thing measured the same way; the open symbols above them are EIRP_{90} , a 90 per cent recovery through the complete automated selection, which is the stricter statement and the one every limit in this paper is quoted on. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) shows the other axis that matters. Archived correlator channels here span 3.3 orders of magnitude in width, from 15.3 kHz to 31.25 MHz, and all of them are far coarser than a dedicated narrowband backend, whose 2.8 Hz channels open the full seven orders only when they are set beside ours. The distinctive feature of this survey is therefore its frequency regime and its channelisation, not its depth. It also samples far fewer systems and far fewer epochs than a dedicated search, and complements those programmes rather than deepening them.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

Because of the very specific and detailed statistical nature of this study, we have extended the paper to include several very detailed Appendices that rigorously explain the methodology involved.

3. THE SAMPLE

This search covers 1651 spectral windows toward 89 stars in 82 systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline, 280 h of telescope time and 1094 h summed over windows. We call that set the *primary census*, and it is the set every limit, every crossing and every expectation in this paper is computed over. Four further sets of blocks enter the paper and none of them is part of it: a hold-out of 77 blocks and 315 windows, reserved before any block on its own side of the rule was searched; an archival extension of 115 blocks and 463 windows, searched afterwards; an out-of-sample control beyond 40 pc; and an external sample used only to calibrate the spatial statistic. Table 6 gives all five sets in rows that sum, with the results each one enters; wherever a number in this paper is computed over something other than the primary census, that table’s name for it is used at the point of use.

Three words are not interchangeable here. A *star* is one stellar component, a *system* a bound group counted once however many of its components were observed, and a *target* one ALMA pointing.

The census is two experiments, not one. 402 windows toward 65 stars in 60 systems are fine-channel, called *Class A* throughout: these resolve a drifting carrier and are the primary experiment, and every sensitivity and every exclusion quoted here refers to them unless stated otherwise. The remaining 1249 windows are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess; they can register a spike but cannot distinguish a drifting signal from a stationary

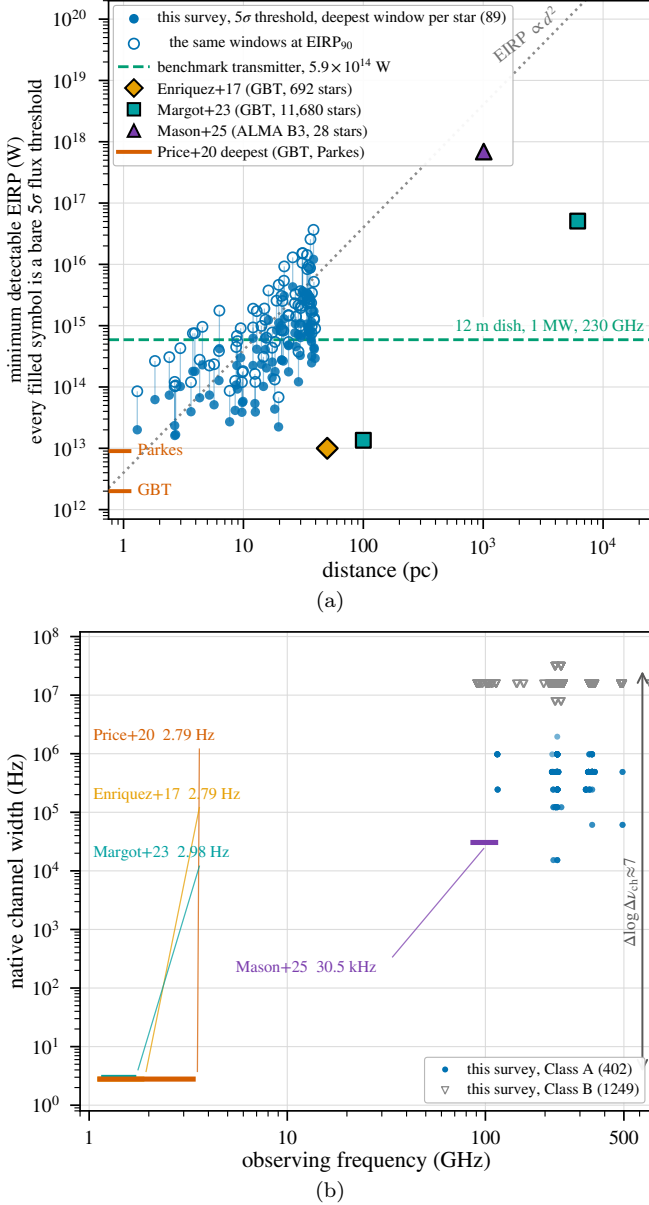


FIG. 1.— Where this survey sits among published radio technosignature searches. (a) Minimum detectable EIRP against distance. Blue, this survey, the deepest window per star: *filled*, that window’s bare 5σ threshold, which is the quantity the other programmes quote; *open*, the same window at EIRP₉₀, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020), all at their own 5σ thresholds. Grey dotted, $\text{EIRP} \propto d^2$ at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of §6.3. The ordinates are commensurable as total powers for a sub-channel carrier, and the filled symbols make the comparison on one convention; the gap to the open symbol is what completeness costs. (b) Native channel width against observing frequency (GHz). This survey’s own widths span 3.3 orders of magnitude; the $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks the span that opens once the comparison programmes’ channels are set beside them: the channelisation, not the plotted power, is what differs most between programmes.

one.

The inclusion criteria are objective and were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. One further rule applies to the window and not to the star: a window is kept only where the primary-beam response at the star is at least 0.5. It withholds 12 windows, all of them ϵ Eridani in Band 6, and no others; Appendix A gives the response of every window kept, and reports limits from the 4 of those 12 that need no beam correction at all. Disc, planet and spectral type enter as descriptive metadata only, and no star or window was excluded by any criterion other than these three, or for short integration. The parallax criterion never binds: the worst fractional parallax error among the stars searched is 2.5 per cent, an order of magnitude inside it and inside the stricter $\sigma_\varpi/\varpi \leq 0.1$ that Table 1 uses to select a reference census on the same footing. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window. The per-star and per-window ledger, including the whole decision tree and every edge case, is in the data release. One target carries no catalogue name and appears throughout under its ALMA source designation: ALMA J153702653–33192492 is Gaia DR3 6015967739554000768, an M5 dwarf at 35.0 pc.

3.1. The archival selection

This sample was not designed. It is whatever previous ALMA programmes happened to observe, for their own reasons, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. The resulting bias is strong and measurable, and Table 1 tabulates it against a volume-limited reference census selected on the same parallax quality.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 49, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Spectral type comes second: A stars are over-represented by about 23 and F and G stars by factors of two to three, because those are the stars with bright debris discs. M dwarfs are mildly depleted — 41 of the 89 searched stars are M dwarfs, a factor 0.8 of their share of the reference census, and only 29 of them carry the fine-channel coverage that can resolve a drifting carrier at all — and that is a weaker statement than the table’s early-type ratios, not a stronger one. Both figures are bounds in the same direction: Gaia temperatures are missing for about half the reference census and missing preferentially for the coolest objects, so the census M fraction is a lower limit and every early-type ratio is therefore a lower limit too. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars, and none of them is an occurrence rate for the solar neighbourhood. The selection acts hardest on exactly the targets a designed search would pick: of the 15 sample systems within 15 pc with

TABLE 1

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS, BOTH SELECTED ON THE SAME PARALLAX QUALITY: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_{\varpi}/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ PC (17 566 OBJECTS, 8594 OF THEM CARRYING A `TEFF_GSPPHOT` ESTIMATE), AGAINST THE 89 STARS WITH AT LEAST ONE SEARCHED WINDOW. THE SEARCHED STARS ARE MEASURED FAR BETTER THAN THE CUT REQUIRES — THE WORST FRACTIONAL PARALLAX ERROR AMONG THEM IS 2.5 PER CENT — SO THE TWO COLUMNS ARE SELECTED ALIKE, AND LOOSENING THE CUT TO 0.2 ADDS NO CENSUS OBJECT AT ALL. 86 OF THE 89 SEARCHED STARS ARE CLASSIFIED: 53 FROM THE GAIA TEMPERATURE, 25 FROM THE SIMBAD SPECTRAL TYPE WHERE THAT TEMPERATURE IS ABSENT, AND 8 IDENTIFIED AS WHITE DWARFS; 3 CARRY NO CLASSIFICATION IN EITHER SOURCE.

THE SIMBAD ROUTE MATTERS BECAUSE `TEFF_GSPPHOT` IS MISSING PREFERENTIALLY FOR COOL STARS, WHICH IS THE POPULATION THE M DWARF STATEMENT IS ABOUT. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION. *The strongest selection effect here is distance, not spectral type.*

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	12 (13.5%)	49.3
5–10 pc	240 (1.4%)	13 (14.6%)	10.7
10–20 pc	2069 (11.8%)	18 (20.2%)	1.7
20–30 pc	5391 (30.7%)	19 (21.3%)	0.7
30–40 pc	9818 (55.9%)	27 (30.3%)	0.5
T_{eff} class ^a			
A	24 (0.3%)	5 (6.4%)	23.0
F	402 (4.7%)	9 (11.5%)	2.5
G	860 (10.0%)	16 (20.5%)	2.0
K	1400 (16.3%)	7 (9.0%)	0.6
M	5908 (68.7%)	41 (52.6%)	0.8
Other axes			
Confirmed planet hosts ^b	26 (15.5%)	23 (25.8%)	1.7
White dwarfs ^a	—	8 (9.0%)	—
Searched stars / systems	—	89 / 82	—

^aTemperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K, normalised over the 78 classified non-degenerate stars. The 8 white dwarfs are kept apart because a degenerate is not an early-type star and its photometric temperature would place it among them. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and every earlier-type ratio in this panel is correspondingly a lower limit; the searched column has no such gap.

^bBoth columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4; a confirmed planet is a row of that table. The 23 searched hosts carry 50 confirmed planets between them, and 21 of the 23 survive the deuterium-burning mass limit. The Gaia reference census carries no planet information, so the census column in this row is the work list rather than the 17 566-object Gaia census.

a confirmed planet, listed in the data release, 6 carry no Class A coverage and so no drift-resolving limit.

3.2. Frequency coverage and archival scope

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: it is 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band. The endpoints are a range over which observations exist, never a range that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §4. The distribution is shown in Fig. 2; a transmitter at a frequency between the islands was never observable here, whatever its power.

The archive is not exhausted either, and the short-fall is accounted for rather than estimated. 177 further

blocks lie inside the same query but were not processed when the census was frozen; Table 6 gives what became of each. 144 have since been calibrated and searched, and 115 of those are reported here, as repeat epochs for the recurrence test and as a sample the frozen pipeline had not seen. The other 29 enter no number anywhere: 13 reached no export, 8 fall below the beam-response floor, and 8 were searched later still. The largest single gap is the 18 blocks for which the archive holds raw visibilities that were never pipeline-calibrated, 237 GB in all; no effort on our side would close it.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. The de-drifted search statistic

Each window’s dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν , and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation — local in time and frequency, global in position, shared by the star and every control, and with the across-position median never subtracted from the numerator. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. T_* denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels

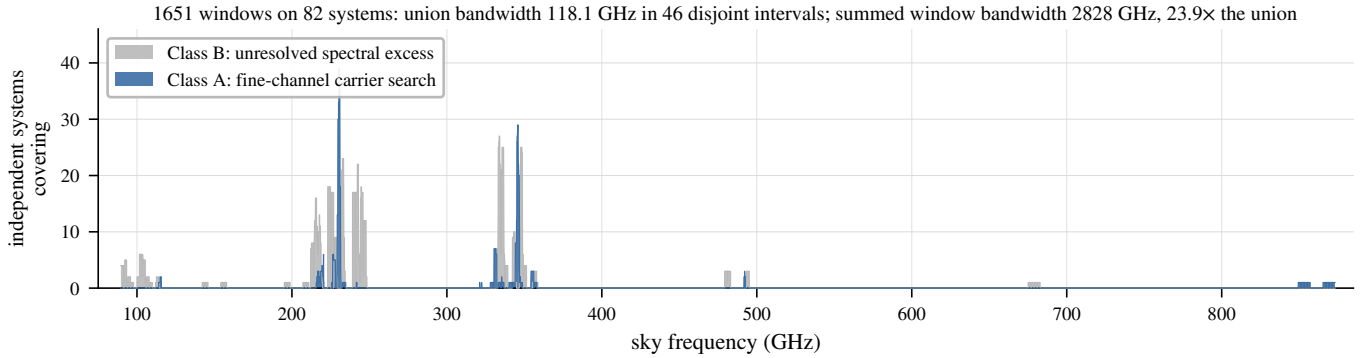


FIG. 2.— Where this survey searched. *Lower panel*: each searched window is a horizontal segment on its host system’s row, rows ordered by distance and labelled alternately on the left and right axes; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel*: the number of systems covering each gigahertz of sky frequency; the threshold distribution is Fig. 1(a). The coverage is narrow and repeatedly revisited: the summed window bandwidth is 23.9× the union, so the union measures frequency coverage and the summed bandwidth search effort.

are smoothed, so a channel’s effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator’s own channel shape supplies the difference, derived and validated against the effective resolutions ALMA reports in Appendix A.

One convention is fixed here and holds everywhere in the paper. The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, since that is what the data contain and what a reader can recompute from the released channel axes. Two steps need the star’s own frame instead, and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block’s frequency axis has been corrected for the Earth’s motion at that epoch and then for the star’s systemic velocity. Nothing here is evaluated in a barycentric frame alone.

Two power quantities carry the main text, both defined in Table 2: the trigger level $P_{\text{trig}} = 4\pi d^2 S \Delta\nu$ at distance d , for a carrier that makes $T_\star$ reach $T_{\text{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90} . The second is measured, not assumed: synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline. Three campaigns did that and they count different things, so each is given with its denominator. The completeness itself is measured on 49 of the 402 fine-channel windows, injected directly, and on 5936 carriers over 35 windows of the coarse class. What the spatial rank costs is measured separately, on 8860 carriers over 52 of 56 attempted block-window configurations. The primary-beam correction is tested on a third set of 40 scored units, 10 of them also among the directly injected windows. Between them they span both ALMA arrays, four bands and the sample’s full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 3. The absolute flux scale is checked twice from outside, in two bands on two arrays: the extractor reproduces an independent reduction of the same AU Mic visibilities to 0.994 ± 0.014 , and the published quiescent flux density of ϵ Eridani to 1.10 ± 0.31 (Burton et al. 2022).

The drift range searched is bounded by an acceleration,

not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units, $a_{\text{los}} = c\dot{\nu}/\nu$, so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band; the released drift and acceleration ceilings satisfy that relation window by window, to 1×10^{-4} over all 402 Class A windows. The default ceiling is $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, that is $|a_{\text{los}}| \leq 3.60\text{--}4.00 \text{ m s}^{-2}$. That is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more. It is far narrower than the general recommendation of Sheikh et al. (2019), which is set by solar-system rotation and orbital motion rather than by a planet’s own acceleration, and the difference is a deliberate restriction of what this search covers and not a weaker version of their advice. The margin is not cosmetic: at unit margin the search would have been blind to two of the windows that produced a crossing. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Two limits follow: the rates searched are the apparent ones at the telescope, so a deliberately chirped carrier is not distinguished from an accelerating platform, and a transmitter need not be bound to a known planet at all, so drifts faster than the ceiling are simply unconstrained (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 2). Channelisation in the sample is bimodal, with an empty gap between the modes, so the split is unambiguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}} \gtrsim 1$. Each window’s own η_{drift} , trial-drift count and acceleration ceiling are released catalogue columns, so a reader may reclassify on that criterion instead; the two do not partition the survey identically. **The two classes are physically different experiments**, results are given per class, and they are combined only where that is statistically appropriate.

4.2. From a crossing to a confirmed signal

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had

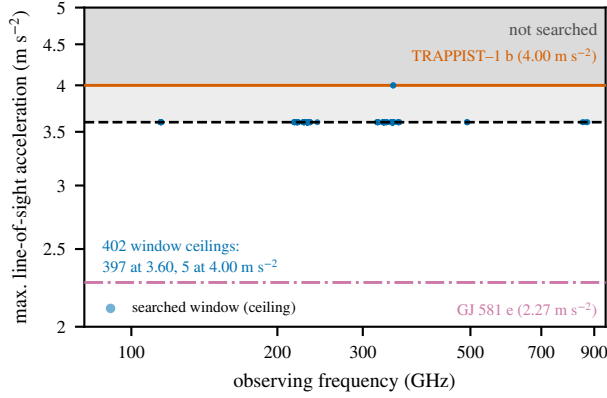


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration, $|a| = c\dot{\nu}/\nu$, against observing frequency; the shaded region above is unsearched. The boundary is flat because the grid was built at constant acceleration; in the units the search works in it is not, running 1.4 kHz s^{-1} at 115 GHz to 10.5 kHz s^{-1} at 872 GHz. Horizontal lines are reference accelerations: Earth’s rotation at the equator (0.034 m s^{-2}) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). The boundary drawn is the default ceiling, 3.60 m s^{-2} , which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST-1 windows. For the most demanding case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling. The widened ceiling falls 0.01 per cent short of that planet’s own maximum acceleration, which leaves 1 per cent of its phase above it.

one survived, a *confirmed signal* (Fig. 7). Each is set out below as the quantity it measures, the number that quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

- (i) *Trigger $\rightarrow$ a crossing.* The quantity is T_* , the de-drifted statistic of Eq. 1 read at the stellar position; the criterion is $T_* \geq T_{\text{trig}} = 5$ in at least one channel $\times$ drift cell; and both the statistic and the level were fixed before the archival search began. **T_{trig} is a trigger threshold and not a detection significance:** T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 of those cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. A crossing establishes only that the window goes on to the next two tests.
- (ii) *Line attribution $\rightarrow$ an unattributed crossing.* The quantity is the crossing’s velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star’s own rest frame; the criterion is $\pm 50 \text{ km s}^{-1}$. The statistic and the trigger were fixed before the archival search began; the line catalogue is a single database query made afterwards, on 2026 October 6, so the dispositions rest on the physical evidence set out beside each one and not on the order in which the list was assembled. The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed* where the species is one that such systems are known

to carry: a coincidence with a catalogued transition makes an astrophysical origin far more likely than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. A further three crossings coincide with a species no debris disc has been shown to carry; their offsets are reported and they are counted as unattributed (Appendix C). A coincidence is in any case a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**, wherever a later block covers the frequency. An unattributed crossing is a target for follow-up and not a detection. The window’s price is that the survey constrains nothing inside it (Appendix C).

- (iii) *Independent recurrence $\rightarrow$ a confirmed signal.* The quantity is T_* at the cell the first epoch predicts, read in another execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive’s uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention first, and neither decides anything. The first ranks T_* against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block’s own synthesised and primary beams so that no control sits on the star; its lowest value, $1/(512 + 1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing’s emission is at the stellar coordinates rather than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank’s own null is called one.

One execution block is not a null, and a rule fixed in advance says so. The comparison with the control positions needs each block’s ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 404 primary-census blocks, 1 fails — A002_Xc7fa6f_X28a8

toward BD+05 1668, at 4.1 where the next worst is 1.12. The Class A results are identical with it and without it, the same 402 windows and 52 crossings either way, because its 4 windows are all coarse-channel; what it does hold is 4 of the 56 crossings, the survey’s entire Class B crossing population (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from the exceedance rate measured at each window’s own control positions; the reserved blocks then test the one assumption that makes, which is that a star behaves like a control position in its own field (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch for proper motion, though not for the annual parallax, whose omission is corrected window by window in §5.1; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the released catalogue. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a measured completeness and not a threshold. Artificial carriers were added to the real calibrated visibilities at a random sub-channel phase and a random drift from each window’s own grid and recovered through the unchanged pipeline; the power at which nine in ten come back is EIRP₉₀, the only sensitivity quoted here. *It is measured through the chain the survey applies* — the trigger and localisation at the stellar position. The 512-position spatial rank annotates a crossing rather than deciding it (§4.2) and is not charged against it: it would cost a further factor 1.31 and in 3 bright-ring windows could not be met at any injected power (Fig. 4), and a limit should not be charged for a gate that disposes of nothing.

Measured that way the completeness is one number. Nine carriers in ten are recovered at $3.03 P_{\text{trig}}$, where P_{trig} is the window’s own trigger power, and that class-average factor is determined to ± 2 per cent. Every one of the 49 directly injected Class A windows resolves a ninety per cent point of its own, between 2.41 and $4.80 P_{\text{trig}}$, and the value does not depend on the brightness of the window’s control ring: the 5 windows whose ring carries resolved emission come back at $3.24 P_{\text{trig}}$ against 2.94 for the rest

TABLE 2
THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER.

Term	Definition
Crossing	A channel $\times$ drift cell at the stellar position reaching $T_{\star} \geq 5$. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than 50 km s^{-1} in the star’s frame from every transition of the line catalogue whose species the system could carry. A target for follow-up; not a detection.
Confirmed signal	A crossing recovered at the predicted stellar-frame frequency and drift in an independent block. The only confirmation criterion, and it is applied to every crossing: one coincident with a catalogued transition would meet it too, and would then be reported as a recurring line.
P_{trig}	The power of a carrier that makes $T_{\star}$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP ₉₀	The power at which nine injected carriers in ten are recovered at the star through the trigger. A measured completeness, not a detection probability.
Class A	The primary experiment: 402 fine-channel windows, median 488.3 kHz, toward 60 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.
Class B	The secondary experiment: 1249 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

(Fig. 4). One factor therefore serves the class. Class B carries its own factor $4.17 P_{\text{trig}}$, measured the same way from 5936 carriers injected into 35 coarse windows of 14 execution blocks and weighted by the coarse catalogue’s own distribution of ring brightness; it is what sets the limits for Barnard’s Star, Wolf 359 and Proxima.

The resulting limits. The median Class A window reaches $2.9 \times 10^{15} \text{ W}$ of equivalent isotropic radiated power and the median system, on its best window, $1.6 \times 10^{15} \text{ W}$, with a spread across systems of 6.8×10^{13} – $3.7 \times 10^{16} \text{ W}$ (Fig. 5a). The round 10^{15} W level is reached for 25 of them. Table 3 collects the terms acting on those numbers. The calibration of the factor is good to $\times 0.91$ – $\times 1.09$, and that is the interval which applies to the median and to the counts. One bias is not inside it and runs one way only: an injected carrier is added to visibilities that have already been calibrated and so suffers no atmospheric decorrelation, whereas a real carrier does. That loss is measured for each execution block from the phase rms the observatory records on its phase calibrator, a point source at the phase centre which loses coherence exactly as a carrier would (Cortes et al. 2024, §10.4.9), through the standard relation between phase scatter and visibility amplitude (Richards et al. 2022),

and it leaves the median window’s limit optimistic by 1 per cent but the median window above 400 GHz by 8 per cent, so the limits at the top of the frequency range are the least secure of those quoted here.

Carriers were injected directly into 49 of the 402 Class A windows, 12 per cent, and the result transferred to the remaining 353, which is the dominant caveat on all of the above. The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance, without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit is the block and not the window*, because a block’s calibrated visibilities are what an injection campaign has to rebuild: band and array were controlled by the draw, while channel width, ring brightness and integration length vary inside a block and were only compared with the population afterwards, agreeing to 22 per cent on the worst axis.

Every limit therefore rests on the class-average calibration, and what one window may differ from it by is a separate statement. The median ninety per cent point differs by 0.18 between bands, 0.24 between channel widths and 0.20 between the two arrays. The largest of the three is the channel width, and none of them is as large as the gate the chain no longer charges for. The array difference survives setting the bright-ring windows aside — 0.20 without them — so it belongs to the array and not to its fields, as a compact array keeping more extended emission inside its beam would. Across the injected windows as a whole the ninety per cent point ranges from $\times 0.80$ to $\times 1.58$ of the adopted factor, so an individual uninjected window’s own completeness is not known to the precision of the interval that applies to the class.

One geometric term acts window by window, and it is measured rather than modelled. The phase centre assumed in the search omits the annual parallax, which displaces the star by a median 0.022 of a synthesised beam and at most 0.67. Each limit is therefore divided by the amplitude a compact source retains at its own displacement — read off an injection into the same visibilities at both positions, not off a beam model (Appendix A) — which makes the median Class A limit shallower by 1.5 per cent. One window, Wolf 28 in A002_X9f2ff8_X25b0, has no astrometric keys and is left uncorrected.

A carrier that is not always radiating is penalised in proportion. The de-drifted stack is coherent across a whole execution block, so a carrier present for a fraction of one contributes to that fraction of the integrations and comes back at that fraction of its continuous amplitude. Over 3888 trials in 12 windows of 10 stars, at 6 duty cycles within a track down to 10 per cent and at amplitudes between $\times 1.8$ and $\times 145$ of each window’s own trigger, the recovered amplitude followed the duty cycle to 1 per cent. A transmitter radiating a tenth of the time therefore needs $10\times$ the power quoted above, and because those trials carry no drift the joint drifting and intermittent case is not measured here at all.

5.2. The nearest stars: Barnard’s Star and Wolf 359

Barnard’s Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-

TABLE 3
WHAT SETS EIRP₉₀, AND IN WHICH DIRECTION. THE UNCERTAINTY ON THE CALIBRATION IS AN UNCERTAINTY ON THE CLASS-AVERAGE FACTOR, AND SO ON ANY STATISTIC FORMED OVER MANY WINDOWS; THE SPREAD BETWEEN WINDOWS IS AN OBSERVED DISPERSION AND IS THE STATEMENT ABOUT ONE WINDOW’S OWN LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR. THE DECORRELATION BIAS IS MEASURED IN 279 OF THE 404 EXECUTION BLOCKS, THOSE WHOSE DELIVERY INCLUDES THE OBSERVATORY’S OWN RECORD OF THE PHASE STABILITY, AND IS NOT IMPUTED TO THE REST.

	value	what it does to the limit
Class-average completeness	$\times 3.03$	nine carriers in ten recovered at this multiple of a window’s own trigger power, over 49 injected Class A windows
Uncertainty on that calibration (per cent), combined in quadrature	± 2	sampling error of the pooled value over the 49 injected windows
The factor itself	± 9	calibration
Flux, distance and visibility scale	$-9 / +9$	$\times 0.91 - \times 1.09$ on the factor, and so on the median limit and on the system counts
Spread between windows. An observed dispersion, not an error bar.		
Window to window	$\times 0.80 - \times 1.58$	the 49 injected windows resolve 2.41–4.80 P_{trig} individually, so an uninjected window’s own completeness is not known to the interval above
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	$\times 1.93$	carried by the injections, so already inside the completeness above
its residual	$+8$	the injected profile falls short of the exact response: conservative
Primary-beam model	$+0 / +10$	ALMA’s beamwidth in place of the pipeline’s Airy-null one: nothing in the median window, 4 windows of one star
Decorrelation	$+0 / +1$	measured block by block from the phase rms the observatory records on each block’s phase calibrator, which an injected carrier does not suffer and a real one does: $+8$ above 400 GHz, and more than $+20$ in 5 per cent of windows

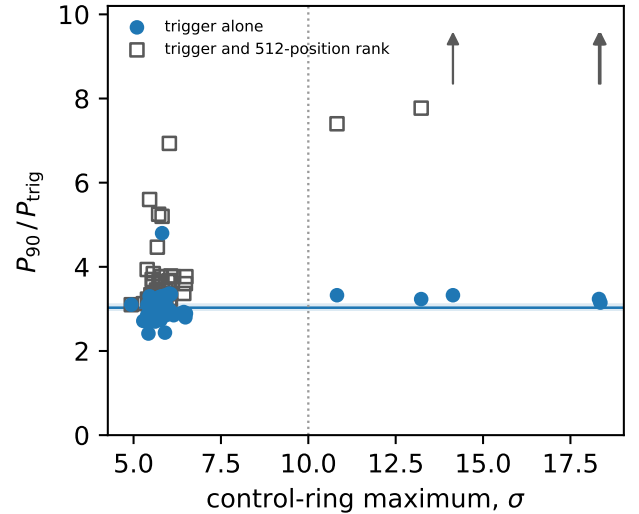


FIG. 4.— Why one completeness factor serves the whole class. Each of the 49 of the 402 Class A windows that were injected directly is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are the criterion adopted here — the trigger and localisation at the star — with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission rather than noise.

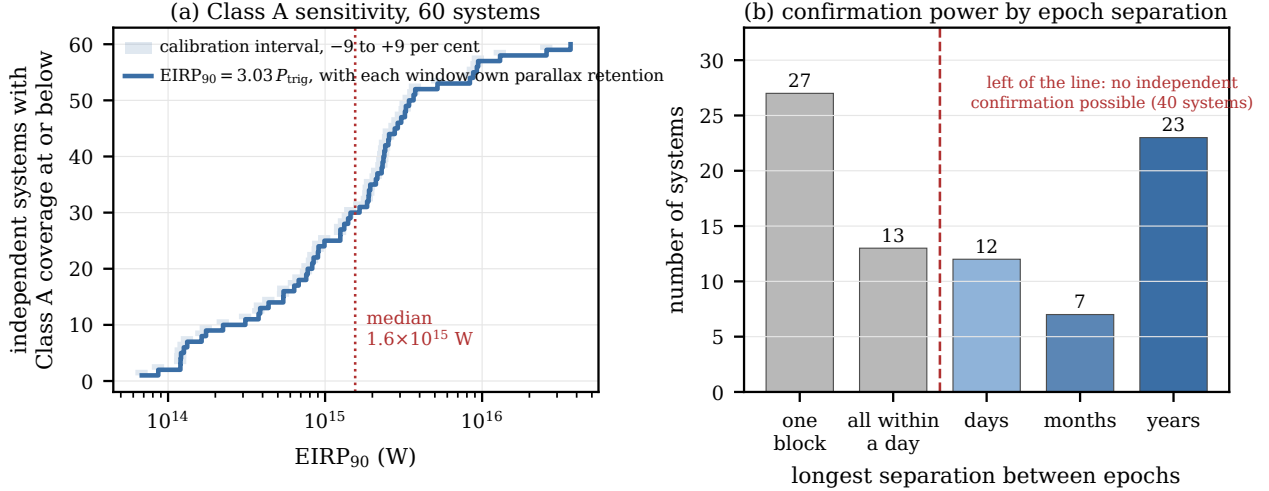


FIG. 5.— (a) Class A sensitivity: the cumulative number of systems whose best window reaches a given EIRP_{90} , the power at which nine carriers in ten are recovered through the trigger at the star. Every window is drawn on the one adopted convention, $3.03 P_{\text{trig}}$ with that window’s own annual-parallax retention applied, and the shaded band is the calibration interval of Table 3. (b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being a separation of more than 1 day. **For the 27 systems observed in a single block, and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a).**

detections. *Both limits come from the secondary Class B experiment and not from the carrier search:* the archive holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the EIRP_{90} convention of §5.1 Barnard’s Star reaches 2.7×10^{14} – 4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14} – 4.5×10^{14} W over 12.

5.3. Threshold crossings and how each was disposed of

The result is a chain of three tests, and it is read in one direction. Of the windows the search covers — the 1651 of the released catalogue and the archival-tail blocks beside them — 56 contain a channel $\times$ drift cell at the stellar position reaching $T \geq 5$, and such a cell is a *threshold crossing*. 4 lie in the one execution block that fails the quality criterion of §5.3.2, fixed before it was applied, so the population characterising the search is the 52 crossings in quality-passing data and those 4 are reported beside it rather than within it. The molecular-line mask, $\pm 50 \text{ km s}^{-1}$ wide and evaluated in each star’s own rest frame, identifies 18 of the 52 with a catalogued transition and leaves 34 unattributed; every crossing is in one column or the other. Of the unattributed, 2 outrank all 512 control positions of their own window, and none was found again in any later epoch covering its frequency. **No crossing survives the chain, and this survey reports no detection.** Figure 7 carries the chain with the number surviving each step and Tables 8 and 9 the crossing-by-crossing ledger behind it.

The mask is informative rather than permissive, and its half-width follows from the Keplerian velocity bound on circumstellar gas in these systems rather than from tuning (Appendix C). The attributions are kinematic and not an accident of bandwidth, and the comparison that shows it is per window and not over the union band: of the 51 crossings for which the release records a measured peak frequency, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected were each peak free to land anywhere on its own window’s channel grid (Appendix E).

5.3.1. Most of the crossings are expected by chance

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells per window spans 4.8 decades, so one level buys very unequal protection. How many crossings chance should produce is therefore measured window by window, over every Class A window of the search and with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute. A window’s own 512 control positions measure how often a position in that field reaches the trigger, and the window’s own channel grid measures how often such a position would fall inside an attribution window, which takes 20 per cent of the total. Chance predicts 29.9 unattributed Class A crossings, with 13–50 allowed by resampling the execution blocks the windows come from, and 34 are found. No window of the search holds more than one crossing, so the two sides of the comparison count the same thing.

The count is consistent with chance, and the width of that range is the honest measure of what the comparison can see: it would not register a small added population in either direction. What tests each crossing individually is the recurrence test of §5.6.

5.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), all of them in one ACA execution block, A002_Xc7fa6f_X28a8, whose whole field is above threshold: 512 of 512 control positions exceed $T = 5$ against a survey median control maximum of 2.75. The single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but the residual does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal — so a slow drift common to every channel and position inflates the statistic across the field, and the 24 sibling executions of the same field, array and spectral setup reach only $T_{\star} = 4.26$ and produce no crossing. *What is inflated is the field and not the star.* The strongest of the four reaches $T_{\star} = 73.4$ where

its own control ensemble peaks at 74.7, and between 1 and 421 of the 512 control positions stand above the star in these windows, so there is nothing here to localise and no evidence either way about a carrier. Applied block by block the same criterion fails this block and no other (Appendix D.2), which is why its 4 crossings are reported apart from the 52 in quality-passing data, and Appendix E takes what the block might instead be to the windows in which the star lies well off the phase centre.

5.3.3. *Stellar flares are excluded by a broadband prediction*

Stellar activity is not excluded by the sample, which contains known flare stars. But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs $49\text{--}4,717\sigma$, against an observed continuum statistic of median -0.29 with $|Z| < 3$ in 45 of them. One block settles it: AU Mic A002_Xa42f75_X319 holds both a crossing ($T_\star = 5.27$) and the published millimetre flare of MacGregor et al. (2020), and *that real, bright, independently published flare contributes 0.17 σ to the single-channel statistic* — raising the continuum at $Z = 10.3$ while the crossing’s own de-drifted track reaches $Z_{\max} = 2.1$.

5.4. *β Pictoris: an astrophysical positive control, and the obstacle it identifies*

A search that finds nothing must show that it could have found something, and the survey’s strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 10 independent execution blocks spanning 8.4 yr, and in 9 of those the star also outranks every control: 5 Band 3 windows at $T_\star$ up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block’s own epoch, then to the star’s rest frame — all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matr   et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block’s own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in an independent execution block, the screened Band 6 window returning $T_\star = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that configuration, which is the geometry the statistic is built to

reject. The rank is therefore a test of compactness and not of authenticity, and on a real extended source it errs conservatively — a stronger statement than a clean positive control, because it bounds the diagnostic’s reach as well as demonstrating it (Fig. 10).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2 $\rightarrow$ 1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block’s $^{13}\text{CO}(2\rightarrow 1)$ window is bright at the ring and faint at the star. What identifies the extended emission is therefore the *level* of the control ring and not the star’s rank against it — the star here outranks every control — which is why the window-level flag of Appendix D.2 is defined on the ring’s median.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 18 of the 56 crossings land on a catalogued transition. The mask that removes them costs 4.3 per cent of the unique-frequency union over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a number on it.

5.5. *The two crossings that lead their own control fields*

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.313811 GHz and CP–72 2713 at 344.269746 GHz, both in Band 7. They are drawn in Fig. 6 and tabulated in Table 4, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4). CP–72 2713’s crossing lies -12 km s^{-1} from SO(8 $_8$ –7 $_7$) in the star’s own frame, a coincidence reported as a note rather than as a disposition (Appendix C). Both events go on to the recurrence test.

Neither returns when the same stellar-frame frequency is observed again. 3 other execution blocks cover 61 Vir’s cell. A carrier persisting at the discovery flux would have given $T_{\text{pers}} = 9.7$ in their combination, which reads $T_{\text{rep}} = +0.39$ at the carried drift, so the two readings of one amplitude disagree by 4.9σ on Eq. B1; searching the whole grid at that cell instead of the one carried drift turns up nothing above 2.71, which is a maximum over every trial and is not a quantity T_{pers} may be set beside. CP–72 2713 is covered by 5 later blocks, the deepest beginning 2.05 hours afterwards and 1.1 times deeper than the discovery; combined they give $T_{\text{pers}} = 6.3$ against -0.82 , or 4.8σ . Appendix B follows that crossing through every step of the chain.

The visibility-domain re-measurement sits beside the chain rather than inside it. 51 of the 56 crossings were re-fitted by phase rotating to the star and following the recovered drift, and 41 are consistent with a point source there; but a noise maximum *at* the stellar position has flux at the stellar position, so at the trigger the fit returns a real part close to the trigger itself, and injected sources in the bin these crossings occupy return $\text{Re}/\sigma = 6.58 \pm 1.97$. Both events are consistent with

a real source and neither excludes one; **the discriminating evidence is the line mask and recurrence.** (The experiment uses three control sets of three sizes: 512 image-plane positions in the rank screen, 8 of them with a retained dynamic spectrum, 12 reachable in the visibilities.)

5.6. Recurrence, applied to every unattributed crossing

A carrier that stays inside the repeat window and radiates through both observations must be there when the same frequency is observed again. That is the confirmation criterion, applied in the star's own frame: the predicted channel follows from the Earth's motion at the repeat epoch and the star's systemic velocity, the discovery drift is carried to it, and the statistic is read there, maximised over nothing. The frame matters: for η Crv it moves the predicted channel by -80.2 channels, so a sky-frequency test would have read a channel holding no information about the event.

One channel assumes more than the data support, because a drifting carrier is a carrier that is accelerating. The drift grid reaches 3.6 ms^{-2} of line-of-sight acceleration, and the largest drift fitted to any of these crossings lies at that ceiling, which is therefore where the bound belongs. A body held there by a star of $2 M_{\odot}$ or less circles it every 4 days at 176 km s^{-1} , so its carrier need not come back to the same stellar-frame channel at all. What is searched at each repeat epoch is therefore the whole spectral window at every drift in the grid, $\pm 833 \text{ km s}^{-1}$ about the predicted channel, which spans that excursion for 30 of the 32 crossings another block covers. Those blocks contribute 328 covering windows, of which 300 hold nothing above the trigger anywhere at any drift; the 28 that do match the 28 chance predicts in trials that size, are already rows of Table 8, and the nearest lies 56 km s^{-1} from a predicted channel.

No crossing was found again. All 38 unattributed crossings were tested, over 444 readings in 151 execution blocks, 42 of them searched after the statistic and the mask were fixed. At the predicted channel the largest statistic over all of them is $+2.00$ at the carried drift and 4.71 over every drift trial, against a trigger of 5. 37 retain a discovery amplitude and so carry an exclusion; 4 of those are withheld, because their amplitude comes from the block that fails the quality criterion of §5.3.2. Over the 33 that remain, the repeat coverage rules out a carrier at a median 0.43 of the discovery amplitude, 19 of them at half that amplitude or less and the deepest at 0.017 of it; but the individual limits run all the way to 1.00, and for 3 of them — β Pic at 241.75 and 241.84 GHz and HD 110411 at 219.45 GHz — nothing below 0.84 of the discovery amplitude is ruled out. As two measurements of one amplitude, discovery and repeat disagree by $3.4\text{--}5.6\sigma$, median 4.8σ : the most the pair can say when the discovery amplitude is itself a five-sigma measurement.

Where the criterion has little power, its silence is not an exclusion. A carrier persisting at the discovery amplitude satisfies it with probability $\Phi(T_{\text{pers}} - 5)$, and one at the trigger amplitude — the faintest that could have produced the crossing at all — with $\Phi(5T_{\text{pers}}/T_{\star} - 5)$, so the $T_{\star}$ and T_{pers} columns of Table 8 give both for every crossing. The first exceeds 90 per cent for 29 of the 33 and runs from 0.41 to 0.85 for β Pic at 241.75 and 241.84 GHz, ALMA J153702653–33192492 at

347.22 GHz and CD–57 1054 at 346.51 GHz, which are untested rather than excluded.

The last crossing, HD 14055 at 331.1559 GHz, retains no discovery spectrum and so carries no exclusion, but the criterion needs only a frequency and a drift grid: 15 of the 17 windows covering it hold nothing above the trigger at all, and the 2 that do hold crossings already in the ledger. Table 8 gives the test crossing by crossing and Appendix B the measured null of the statistic read there.

5.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements, and this experiment has two extents. It searched 60 systems and 402 Class A windows over 47.7 GHz of union bandwidth; it could have *confirmed* a signal in 30 of those systems, the ones observed again at the same frequency more than a day apart, over 23.3 GHz or 49 per cent of that union, and in 11 systems over 8.5 GHz beyond a year (Fig. 8). The difference is the cost of retuning.

An independent execution block is not an independent epoch. Every test above uses a block other than the discovery block, but the blocks are not all separate in time: the usable baselines run from 0.46 hours to 1.8 yr, and for 21 of the 37 crossings the nearest covering block falls within a day, while 1 have no covering block beyond one. What these tests exclude is a transmitter whose carrier stays inside the searched window and radiates through both observations: over the hours to days where there is repeat coverage they constrain that, over months and years they bound no duty cycle, and for the 44 systems with no second block covering a frequency the first one covered they bound nothing at all. The null reported here is a statement about transmitters that would have been seen again.

5.6.2. The reserved hold-out and the control beyond 40 pc

Two sets of execution blocks were searched with the frozen pipeline and then reserved for calibration (Table 6); both hold data on real targets and both are reported here as searches. The hold-out — 77 blocks, 315 windows, 36 stars and 214 h on source, withheld by a rule fixed before the search finished and sharing no execution block with the census (Appendix B.1) — contains 8 threshold crossings, the strongest at $T_{\star} = 8.64$, and *not one outranks its own control positions*: the fewest controls above the star in any of them is 5 of 512. 7 were tested for recurrence on the same code path as the census, and exactly one is present again: HD 285968 at $+9 \text{ km s}^{-1}$ from CO(2–1), whose 4 later blocks combine to $T_{\text{rep}} = +6.71$ at the carried drift, the value Table 5 prints, and reach 5.87 when the drift is maximised. It is a molecular line, and *that* it recurs is the test working in the positive direction; for every other tested crossing the repeat coverage rules out a carrier at 0.47 of the discovery amplitude or less (Table 5).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88 h on source, with 8 crossings whose strongest is $T_{\star} = 12.7$ and none of which outranks its own controls. A systemic velocity is known for only 0 of them, so it supports no limit and shows only that the pipeline behaves the same way on fields it

TABLE 4

THE TWO EVENTS OF FIG. 6. THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. REPEAT EPOCHS MEASURED ARE THE BLOCKS AT WHICH THE PREDICTED STELLAR-FRAME CELL WAS READ; THE INDENTED ROWS GIVE HOW MANY OF THOSE THE RELEASED CATALOGUE CARRIES AND HOW MANY BLOCKS IT LISTS AS COVERING THE FREQUENCY AT ALL, WHICH DIFFER BECAUSE THE ARCHIVAL EXTENSION IS DEPOSITED SEPARATELY. T_{rep} COMBINES THE REPEAT EPOCHS BY INVERSE VARIANCE AT THE CARRIED DRIFT: IT IS ONE CELL AND ONE TRIAL, SO IT MAY BE NEGATIVE, AND IT IS THE QUANTITY T_{pers} IS COMPARED AGAINST. THE INDENTED READINGS BENEATH IT ARE THE LARGEST SINGLE EPOCH AND THE LARGEST OVER EVERY DRIFT TRIAL, EACH A MAXIMUM AND SO POSITIVE ON AVERAGE WHATEVER IS THERE. THE EXCLUSION IS EQ. B1. BENEATH IT ARE THE FRACTION OF THE DISCOVERY AMPLITUDE S STILL RULED OUT AT THE TRIGGER, $(T_{\text{rep}} + 5)/T_{\text{pers}}$, AND THE PROBABILITY $\Phi(T_{\text{pers}} - 5)$ THAT A CARRIER PERSISTING AT S , AND THEN AT THE TRIGGER AMPLITUDE $5S/T_*$, WOULD HAVE SATISFIED THE CRITERION AT ALL. THE POWER QUOTED IS THAT IMPLIED BY THE FLUX IN THE PEAK CHANNEL; AN UNRESOLVED CARRIER LEAVES ONLY 0.48 OF ITS POWER THERE, SO THE POWER ACTUALLY RADIATED IS $\times 2.00$ – $\times 2.36$ LARGER, A FACTOR ALREADY INSIDE EIRP₉₀ AND NOT INSIDE THIS ROW.

	61 Vir	CP-72 2713
Frequency (GHz)	346.313811	344.269746
Channel width (kHz)	488.28	488.28
Drift rate (Hz s ⁻¹)	+3182	-2056
T_*	5.958	5.809
Highest control T	5.65	5.68
Flux density (mJy)	118.8 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.1 × 10 ¹⁴	8.9 × 10 ¹⁴
Repeat epochs measured	3	5
in the released catalogue	3	1
catalogue blocks at ν	4	1
Observed	2017 May 20	2022 Oct 2
Nearest covering epoch	2017 May 19	2022 Oct 2
separation (h)	-1.97	+2.05
covering epochs more than a day away	2	4
Longest baseline (d)	3	2
T expected if persistent, T_{pers}	9.70	6.30
T measured, matched drift, T_{rep}	+0.39	-0.82
largest single epoch	+1.65	+0.96
maximised over every drift	2.71	3.60
Exclusion, σ_{excl}	4.9 σ	4.8 σ
Amplitude still excluded at 5 σ	0.56 S	0.66 S
Recovery chance if persistent, at S / at 5 σ	1.00 / 1.00	0.90 / 0.66
Disposition	not confirmed	not confirmed

was not tuned for. The third reserved set calibrates the spatial rank and is not external at all (Appendix B).

5.7. A stellar-frame multi-epoch stacked search

Combining the epochs of a star in that star’s own rest frame deepens the limit. 55 of the 82 surveyed systems hold more than one searched execution block, and 58 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 315 stacked groups, 2177 epoch-windows and 1426 h, the flux limit improves by a median factor 1.52 over each star’s best single epoch and 4.45 at most. 11 further stars stack too — 10 control stars beyond 40 pc and 1 covered only by archival blocks — and are excluded from those figures; over all 69 stacked stars, whose stacks are mostly shallow two-epoch ones, the median gain is 1.46. Every stacked limit is measured by injection into the stack itself and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a me-

dian 8.2 km s⁻¹ between the epochs of one star and by as much as 53.9 — many channels at the 488 kHz Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star’s catalogued identity rather than the directory the search ran in. Spectra with pathological archived visibility weights are excluded first (27 of 2838, which would otherwise dominate the sum). The transform is verified on a real line: HD 285968’s molecular line collapses from a 19.26-channel topocentric spread to 0.798 channels over 4 epochs spanning 44 d.

The criterion was fixed before any stellar position was examined: a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, measuring the whole test’s false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is $\sqrt{N_{\text{eff}}}$ and not $\sqrt{N}$, where $N_{\text{eff}} = \sigma_{\text{min}}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median N_{eff}/N is 0.65. What a stack realises is $\sqrt{N_{\text{eff}}}$ divided by the measured noise of its own stacked spectrum, and that noise follows the inverse-variance prediction to 0.992 in the median, with a per-stack scatter of 7 per cent and nine stacks in ten between 0.81 and 1.18. Against $\sqrt{N}$ the realised gain is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads. HD161868 stacks 13 epochs for a gain of only 1.15: its N_{eff} is 1.7, so $\sqrt{N_{\text{eff}}}$ is 1.29, and its stacked spectrum came out 1.12 times as noisy as predicted.

The deepest stacked limit is Proxima Centauri’s, and it is a Class B limit. Proxima is covered by 66 blocks over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. 25 of those blocks belong to the primary census, 33 were searched afterwards from the same public archive and 8 come from the reserved hold-out, all observed over the census blocks’ own years. Stacking the census blocks alone, the set every other limit here uses, gives EIRP₉₀ = 2.2×10^{13} W; adding the rest of the archive deepens that by 1.26 to 1.73×10^{13} W, against 5.51×10^{13} W for the best single epoch on the same injection-calibrated convention. Its N_{eff} is only 7.9, so $\sqrt{N_{\text{eff}}} = 2.80$ of the measured 3.19 is what combining the epochs buys, the rest being this spectrum’s own noise. The limit holds for an emitter steady in the stellar frame to within one 15.625 MHz channel, 20.9 km s⁻¹ peak to peak, with no inter-epoch drift above 7.8×10^{-2} Hz s⁻¹. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known lines. They are β Pictoris’ CO $J = 1-0$ at +3.09 km s⁻¹ over 8 epochs ($Z = 45.3$) and its $J = 2-1$ at +3.55 km s⁻¹ ($Z = 12.1$), agreeing to 0.46 km s⁻¹ or 0.73 of a CO(1-0) channel: the astrophysical positive control passing in the stack. The survey’s own ± 50 km s⁻¹ mask leaves 0 of them unattributed. HD 285968 reaches $Z = 12.89$ on the line used above as the registration standard and is rejected by its own controls at 14.19: the emission is

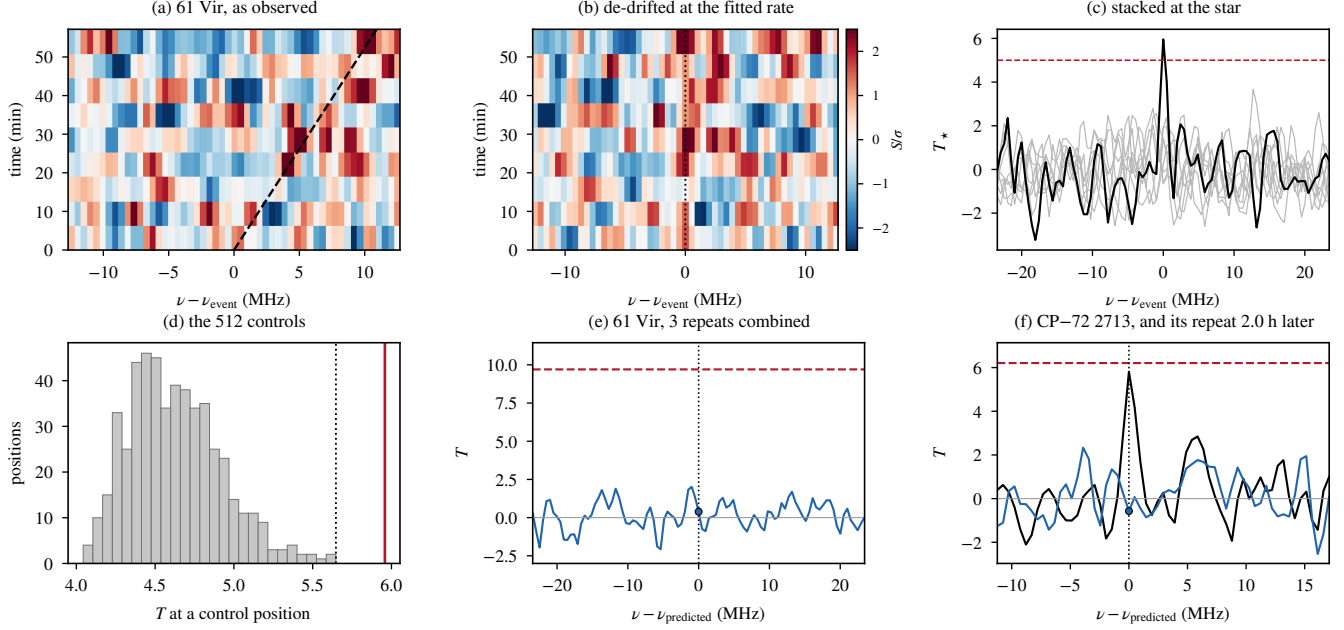


FIG. 6.— **Why this survey reports no confirmed signal although it produced an interesting event.** (a) The dynamic spectrum at 61 Vir’s position around the crossing at 346.313811 GHz, observed 2017 May 20, with the fitted drift of $+3182 \text{ Hz s}^{-1}$ as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin’s noise. At $119 \pm 20 \text{ mJy}$ against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them: it exists only in the coherent stack of all 38 minutes on source. (b) The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the 57 minutes the observation spans; 8 of the 9 bins are positive at the marked channel. (c) The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. (d) The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted). (e) The 3 other blocks covering that cell in the star’s rest frame, combined, with zero at the predicted channel, which the frame transport moves between epochs (Table 4 dates them). Dashed, the $T_{\text{pers}} = 9.7$ a carrier persisting at the discovery flux would have given; circled, the $T_{\text{rep}} = +0.39$ read there at the carried drift, 4.9σ away on Eq. B1. (f) The second event, CP-72 2713 at 344.269746 GHz : its discovery stack (black) and the same cell 2.05 hours later in a block 1.1 times deeper (blue), with the same two marks at $T_{\text{pers}} = 6.2$ and $T_{\text{rep}} = -0.57$. The two events imply isotropic-equivalent powers of $5.1 \times 10^{14} \text{ W}$ at 8.5 pc and $8.9 \times 10^{14} \text{ W}$ at 36.7 pc .

TABLE 5

THE THRESHOLD CROSSINGS OF THE RESERVED HOLD-OUT, SEARCHED WITH THE SAME FROZEN PIPELINE AND DISPOSED OF BY THE SAME TESTS AS THE CENSUS. Δv_* IS THE OFFSET FROM THE NEAREST MASKED TRANSITION IN THE STAR’S OWN FRAME AND n_{rep} THE NUMBER OF COVERING REPEAT BLOCKS; T_{pers} AND T_{rep} ARE AS IN TABLE 8. THE ONE CROSSING THAT IS PRESENT AGAIN IS A CARBON MONOXIDE LINE.

Star	block	ν_{topo} (GHz)	T_* line	Δv_*	n_{rep}	T_{pers}	T_{rep}	disposition
HD 285968	Xdaef5a_Xdd9b	230.5081	8.642 CO(2–1)	+8.7	4	11.6	+6.71	attributed; present again
HD 14055	X107376c_X6126	330.7123	5.513 $^{13}\text{CO}(3-2)$	+112.6	30	312.0	+0.56	unattributed; does not recur
TRAPPIST-1	Xd0e450_X259b	346.2454	5.326 SO(9_8-8_7)	-306.6	5	11.1	-0.32	unattributed; does not recur
HD 172555	X107fa96_X1300f	344.6066	5.144 SO(8_8-7_7)	+251.3	—	—	—	unattributed; no repeat coverage
HD 285968	Xdb4b9a_Xa7c	229.9129	5.106 CO(2–1)	-768.0	4	11.2	+0.21	unattributed; does not recur
HR 1010	Xc6b674_X23f2	346.2301	5.044 SO(9_8-8_7)	-238.7	7	25.9	-0.80	unattributed; does not recur
HD 10647	Xff99e1_X9cdf	345.7059	5.017 SO(2_3-2_1)	+37.1	13	142.1	-0.93	attributed; does not recur
HD 170773	Xff294f_X1fb	345.5597	5.001 SO(2_3-2_1)	-113.8	4	59.3	+1.45	unattributed; does not recur

extended, as a cloud is and a transmitter is not.

The restriction travels with every number above, and its width is the channel. At CO(2–1) one 488 kHz channel spans 0.63 km s^{-1} , the whole excursion a stacked carrier is allowed, peak to peak. A transmitter on a short-period giant planet, its line-of-sight velocity swinging by $\pm 30 \text{ km s}^{-1}$, crosses 94 of those channels peak to peak and de-registers completely, against 3.0 of a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way, a stack in 488 kHz channels being confined to $1.3 \times 10^{-1} \text{ Hz s}^{-1}$ over the median inter-epoch span of 44 d. No crossing of §5.6 drifts that slowly: the slowest measured is 519 Hz s^{-1} , which leaves one channel in 16 minutes, and in 33 of 38 cases the stack covering a crossing contains its

discovery block. Toward the 20 stars carrying them the stack is a deeper limit on a non-drifting carrier, not a test of these events. Each stacked limit therefore constrains only emitters stable in the stellar frame to within its own channel; the single-epoch limits of §6.3 apply generally.

6. DISCUSSION

6.1. What is different about a millimetre technosignature search

The most transferable result of this experiment is not the non-detection but a measurement of what limits a carrier search at these wavelengths. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity. That is a statement about disposing of events and not about depth:

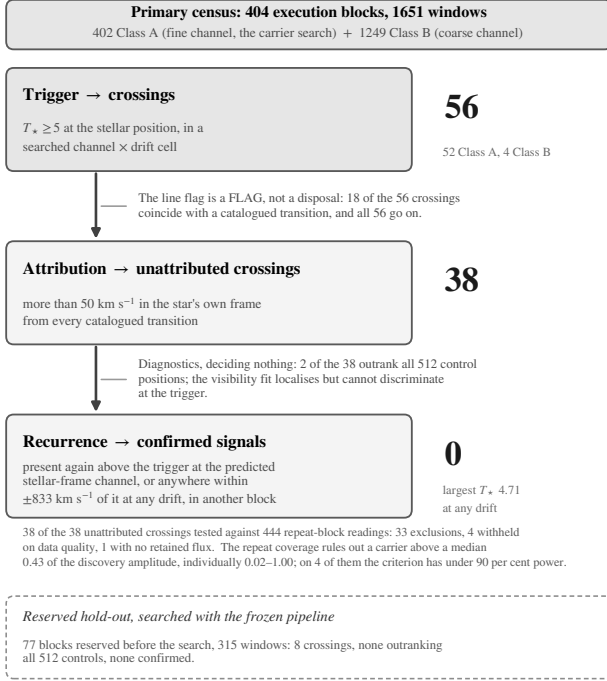


FIG. 7.— The detection logic, with the count at each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger and the catalogue, so the figure, Table 8 and the text cannot disagree.

the power a transmitter must radiate to be found here remains a median $1.6 \times 10^{15} \text{ W}$ per system.

Confusion, and not interference, sets the false-positive rate. Catalogued molecular transitions, almost all of them carbon monoxide, account for 18 of the 56 threshold crossings, and no crossing in this survey is attributed to interference (Appendix E). 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, arising in 2 of the 69 debris-disc systems searched. The remedy that works is a velocity-space mask evaluated in the stellar frame, and it is not free: it removes 4.3 per cent of otherwise usable coverage.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its resolved disc emission lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is $\times 2.464$ the interior rate; without that window $\times 1.020$, and on a held-out half $\times 1.181$ ($p = 0.24$). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

The channelisation is inherited. 1249 of the 1651 windows have channels so coarse that every drift in the grid

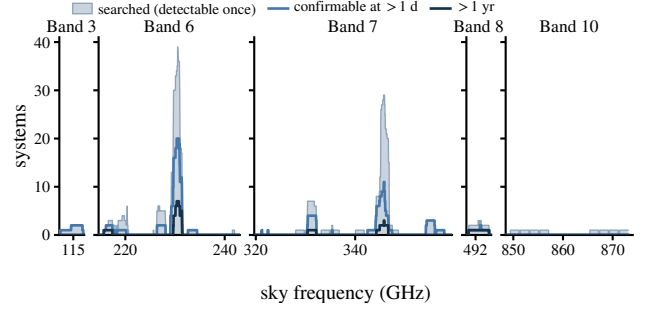


FIG. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9.

stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 402 of them. The widths on offer here run from 15.3 kHz to 31.25 MHz, about 3.3 orders of magnitude; set beside the 2.8 Hz channels of centimetre-band carrier searches the range becomes 7 orders (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\text{conf}}(\nu)$ be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency ν , so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 23.3 GHz of union bandwidth, 49 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 20 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

Finally, a threshold is not a sensitivity. A trigger level is a property of the noise, not of what a transmitter of a given power does to a pipeline: measured by injection and recovery through the unchanged selection, the power needed for ninety per cent recovery is 3.03 times a Class A window's own trigger power and 4.17 times a Class B window's.

6.2. The search space this archive opens

The carrier search is 402 fine-channel windows toward 60 stellar systems, 47.7 GHz of union bandwidth; the 1249 coarse-channel windows are a supplementary spectral-excess experiment. No previous search has looked for technosignatures toward stars within 40 pc above about 30 GHz, and the one earlier millimetre search worked in Band 3 toward far more distant stars (§2); Band 3 is searched here, in 112 windows down to 90 GHz including β Pictoris' CO $J = 1-0$ control. Of the 10 MHz–115 GHz haystack axis of Wright et al. (2018) this experiment covers 19.9 GHz, only 1.0 GHz of it car-

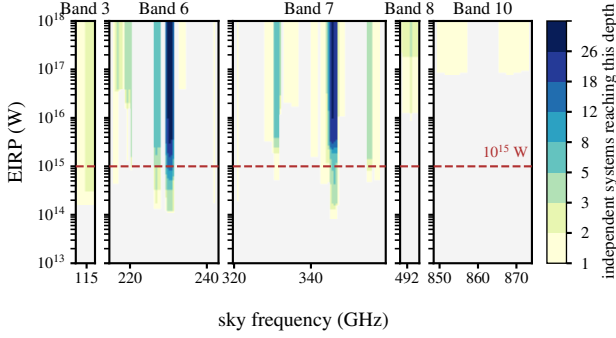


FIG. 9.— Where the experiment could have found something. Shading gives the number of independent systems that would have yielded a recovery at 90 per cent completeness for an unresolved carrier of the plotted EIRP at the plotted sky frequency; unshaded means none. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands across 114.0–873.1 GHz, which a continuous abscissa would draw as hairlines. The dashed line is 1×10^{15} W. Figure 2 shows where ALMA looked; this shows where the experiment was sensitive.

rier search because the Class A union begins at 114 GHz; the rest lies above that axis.

Union bandwidth overstates what that buys: different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 gives instead the number of independent systems that would have yielded a recovery at ninety per cent completeness, against sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems and 17 at the best-covered frequency near 231 GHz; with no limit on power that frequency reaches 39 of the 60 systems, and half the sample only above 3.4×10^{15} W.

Archival SETI buys sky and frequency cheaply and buys confirmation not at all: 82 systems and 118.1 GHz of union bandwidth across both classes without an allocation, on data selected by others and so carrying the biases of §3.1.

The conventional measure for comparing programmes is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), $\text{CWTFM} = \zeta_{\text{AO}} \text{EIRP}_{\text{min}} / (N_{\star} \nu_{\text{rel}})$ with $\nu_{\text{rel}} = \Delta\nu / \nu_{\text{mid}}$, smaller being better. Those programmes quote EIRP_{min} at a bare detection threshold, so the Class A experiment is put on that footing: with $N_{\star} = 60$ and $\nu_{\text{rel}} = 0.097$ it is 1.0×10^5 on the sample’s largest threshold power and 4.4×10^3 on the median system, against 0.85 for that survey. Margot et al. (2023) publish no CWTFM; the same expression on the three quantities they do give — 11 680 stars, 1.15–1.73 GHz and $\text{EIRP} > 1 \times 10^{13}$ W — returns 0.11. With the EIRP_{90} adopted everywhere else here the same two become 3.2×10^5 and 1.3×10^4 . Nearly all of the difference is the product $N_{\star} \nu_{\text{rel}}$: this is a far smaller haystack than a wideband centimetre survey of thousands of stars, and the measure compares surveys rather than estimating occurrence rates.

6.3. What this experiment excludes, and what it does not

Excluded. Consider a continuously transmitting, unresolved carrier that lies outside the molecular-line mask, inside the searched frequency range and inside the linear-

drift grid, and that is present during one of the archival epochs. Such a carrier would have been found nine times in ten above the EIRP_{90} of §5.1, a median of 1.6×10^{15} W per system on its best window. None was detected toward any of the 60 systems the carrier search covers, at that recovery probability and above that limit.

Four further axes bound the excluded population as sharply as power does. *Frequency:* the coverage is sparse and inherited (Fig. 9) and the median system is covered over only 1.84 GHz. *Drift and acceleration:* the grid reaches a line-of-sight acceleration of $3.60\text{--}4.00 \text{ m s}^{-2}$, which contains every known planet of the sample but one (Fig. 3); an oscillator drifting faster than that lies outside it. *Persistence:* the carrier must be radiating during one of the archival observations. *Spectral width:* the statistic is built for emission confined to one 15.3–1953 kHz channel, and anything broad enough to resemble continuum is removed with the spectral baseline.

Not excluded. A transmitter absent during the archival observation, or on for much less than the length of one, at the cost in power §5.1 measures. Emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, at any strength, which costs 5.04 GHz of union bandwidth. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below a window’s own completeness. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

An illustrative engineering benchmark. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D \nu / c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, a depth this survey reaches for 16 of the 60 Class A systems. The same gain fixes how narrow the beam is, and detection requires the Earth to lie inside it: a solid angle $\Omega = 4\pi / G = 2.1 \times 10^{-8} \text{ sr}$, about 34 arcsec across, illuminating a fraction $\Omega / 4\pi = 1.7 \times 10^{-9}$ of the sky at a time. The example shows that EIRP is not transmitter power — the same megawatt radiated isotropically would be 10^6 W — so these limits apply to beamed transmitters whose beam contains the Solar System and say almost nothing about omnidirectional beacons.

Two blind spots. The search is Stokes *I* only, so the discrimination that polarisation would offer against unpolarised astrophysical emission is unavailable; ALMA’s feeds are linear, and the circularity of a circularly polarised carrier lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s and not a choice: none of the “magic frequency” arguments that have guided centimetre-band searches falls in an ALMA band, so a transmitter deliberately placed on one lies outside this experiment by construction.

6.4. What a dedicated ALMA technosignature experiment should do differently

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere:* the 1249 coarse-channel windows cannot carry a carrier search at all. *Choose the targets and the frequencies:* an archive weighted toward dusty discs is the

worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 5.04 GHz the mask costs — all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning*: two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities*: rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. No published radio technosignature search above about 30 GHz has reached a star within 40 pc (§2), and an archive assembled for other science is at present the only way to reach one.
2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of union bandwidth in disconnected intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1249 windows toward 81 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform reduction of 404 public execution blocks, 4.1 TB of raw data, and between them they span 90–873 GHz.
3. Sensitivity is a measured completeness, not a threshold: EIRP₉₀ is the power at which nine of ten carriers injected into the real calibrated visibilities come back through the unchanged selection, the trigger and localisation at the star. The median system's best window reaches 1.6×10^{15} W, with a spread across systems of 6.8×10^{13} – 3.7×10^{16} W. The rank and the visibility fit decide nothing, so neither is charged to that limit. One bias runs one way only and is not inside the quoted calibration: an injected carrier suffers no atmospheric decorrelation where a real one would, and the coherence each block actually lost, measured on its own phase calibrator, makes the median limit optimistic by 1 per cent and those above 400 GHz by 8 per cent.
4. All 56 threshold crossings are reported with their dispositions, four of them in the one execution block that fails the data-quality criterion of Appendix D.2 and reported apart from the 52 in quality-passing data; of those, 18 coincide with a catalogued molecular transition, almost all of them carbon monoxide — including the β Pictoris disc, which the search recovered without being told where to look — and 34 do not. Of the unattributed, three coincide with sulphur monoxide, which no debris disc is known to carry and which here would have to outshine an undetected carbon monoxide line in the same window; the coincidence is reported and not acted on, and disposing of them instead would make the split over all 56 crossings 21 to 35 rather than 18 to 38. Chance, measured from each window's own control positions and channel grid, predicts 29.9 unattributed Class A crossings with 13–50 allowed by resampling the execution blocks, so the count is consistent with chance within a range that would not register a small added population either way.
5. Every unattributed crossing was tested for recurrence in the star's own frame, in each further execution block covering the event frequency and over the whole of each such window at every drift searched, because an accelerating carrier need not return to one channel. None was found again: the largest statistic at any predicted cell is 4.71 against $T_{\text{trig}} = 5$, and the exceedances found anywhere in those windows are no more than chance predicts for trials of that size. Nor did the two that lead their own control fields return. Of the 33 crossings that retain a discovery amplitude and so carry an exclusion, the repeat coverage rules out a carrier at a median 0.43 of that amplitude, with the individual limits running from 0.017 to 1.00: the median is not a bound on the set. Where the repeat is shallow the criterion has little power and its silence is not an exclusion — a carrier radiating at the discovery amplitude through both observations would have met it with probability above 90 per cent for 29 of those crossings but only 0.41–0.85 for the remaining 4, which are untested rather than excluded. What these tests exclude is a carrier that holds its frequency inside the window searched and radiates through both observations, and not a persistent transmitter of any kind.
6. What this survey could search and what it could confirm are different extents: of the 60 systems the carrier search covered, only 30 were observed again at the same frequency more than a day apart and 11 beyond a year. Across both classes, 44 of the 82 systems have no second block covering a frequency the first one covered, and toward those the null reported here says nothing at any strength.
7. Combining each surveyed star's epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri, where $\sqrt{N_{\text{eff}}} = 2.80$ is what combining the epochs buys, the stacked limit measured by injection into the stack reaches 2.2×10^{13} W over the primary census alone and 1.73×10^{13} W once the rest of the archive is added, in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to within one channel, 20.9 km s^{-1} peak to peak, across 6.3 yr.
8. These results constrain unresolved carriers only within the domain actually searched and only toward this archive-selected sample (§6.3); what transfers beyond it is the obstacle, that a channelisation chosen by other observers for other purposes, and not collecting area, decides which kind of search is possible at all.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and pro-

duced the released catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table 8.

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COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 June 11 and belong to 65 ALMA proposals, whose codes, in the standard ADS/JAO.ALMA#*code* form, are recorded block by block in the data release together with each block’s epoch. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France), of the NASA Exoplanet Archive (Akeson et al. 2013), and of the Splatalogue spectral-line database (Remijan et al. 2007).

DATA AVAILABILITY

The reference of record is a single permanent Zenodo deposit, cited by its DOI. It carries every number in this paper and the code that produced them, and contains: the 89-star target catalogue with its selection function; one row for each of the 1651 searched windows, carrying its execution-block identifier, frequency limits, native channel width, trial-drift grid, trigger statistic at the stellar position, control statistics of the 512-position ring and the window’s EIRP₉₀; the ALMA proposal code and epoch of every execution block, so that each of those rows resolves to the programme that took it; all 56 threshold crossings with their stellar-frame frequencies, line classifications and recurrence measurements, including for each the probability that a carrier at the discovery and at the trigger amplitude would have satisfied the confirmation criterion, and the catalogue’s own disposition column as computed at extraction — in the observed frame, on the first extraction’s statistic and against the search-time line list, none of which this paper adopts, so Table 8 is the authority for every disposition quoted here; the search statistic and the molecular-line list as files in their own

right; the injection–recovery records; the block-by-block archival verdicts and the ALMA delivery record for every block in scope that could not be searched; the diagnostic figures the printed paper refers to but does not reproduce, among them the per-system coverage waterfall and the selection funnel (`coverage_waterfall_full.pdf`, `selection_funnel.pdf`); the illustrative table of sample systems within 15 pc with a confirmed planet; and the full pipeline with the commit that produced the deposited products. Two properties of the deposit are measured rather than claimed: 148 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline.

Three of the choices that could create or remove an unattributed crossing carry an identifier that can be checked independently of the deposit. The calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were committed to the project repository as `c75069040eab52ec1336376a4dae62114bb281a3` on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The molecular-line list is a single dated query of a public line catalogue (2026-10-06) deposited as a file, so it is reproducible rather than described; it post-dates the fixing of the disposition criteria on 2026 September 11 (`f5f52e7804f1baad99515ea2d46ddd7e4e4cd546`), and so does every species in it, so no coincidence is set aside on the ground that its species was recognised late and none is privileged by having been recognised early. Where a coincidence is reported and not acted on, the reason is physical and applies to every coincidence of that species (Appendix C). The repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

Checking a threshold in this paper needs three things: the frame every frequency is quoted in, the instrumental response that separates a channel threshold from a transmitter power, and the primary-beam factor that turns a measured noise into a flux at the star. The per-window catalogue, `catalogue/windows.csv` in the deposit, carries the columns each of them uses.

Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search itself applies no velocity-frame shift. Velocities enter at

the vetting stage, where the line mask is evaluated in each star’s own rest frame: it carries every crossing frequency to the barycentre at its own block’s epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable. One number fixes the scale, illustratively and not achievably: rechannelising the median Class A window’s 488 kHz to the 2.79 Hz of a dedicated narrowband search would divide the threshold by 418, and no reprocessing recovers resolution the correlator never recorded.

Channel widths come from the measurement sets’ own `CHAN_WIDTH`; MS v2 defines no “`EFFECTIVE_BANDWIDTH`” column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2024), so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak channel holds 0.50 of an unresolved carrier’s flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase, so a nominal threshold is optimistic by $\times 2.00$ – $\times 2.36$; the released catalogue carries the nominal threshold and both corrected values for every window.

The injected carriers are deposited with the correlator’s own channel response, at a sub-channel phase drawn afresh for each tone, so the smoothing loss sits inside the measured completeness rather than being applied to it. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much. Where in a channel a carrier was placed makes no difference to its recovered signal-to-noise, because a carrier at a typical searched drift rate crosses about 11 channels during one observation and so samples the whole response. Since the loss is carried end to end it must not be applied to the result as well: at the ninety per cent point of the recovery curve the peak channel holds $1.93\times$ the nominal trigger flux, which is that factor of about two, already spent.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage lo-

cal null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so the correlation is instrumental and those spectra are smoothed and un-averaged; and the archive’s reported effective spectral resolution is exactly twice the channel separation in 1505 of the 1651 windows, in both correlator modes. The other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller; every released window has a reported resolution, so the blanket factor is right for the great majority of both classes. *The correlation model is truncated at lag two:* the baseline step, a per-integration median over up to 65 channels, could imprint longer-range correlation that the released products retain no residuals to measure, which would make the independent-trials count of Appendix D an over-estimate.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is $\approx 32 \text{ MHz}$ at 488 kHz and $\approx 0.9 \text{ GHz}$ at 15.62 MHz. Within that ceiling a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so its threshold is N_{ch} times the single-channel one, which is the rescaling the main text applies to signals broader than one channel.

Intra-integration smearing, which the injections do not model. A drift smeared by Δ channels within one integration keeps $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of the peak amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smear}}$. Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one, and this term has to be evaluated rather than measured. It can be evaluated wherever the release carries the integration count, which is 451 of the 1651 windows; over those the excursion reaches at most 1.10 channels, only 9 have $\eta_{\text{smear}} < 0.99$ — HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP–72 2713 Band 6 (0.979, $\times 1.02$) — and the other 442 agree with their nominal thresholds to better than one per cent. Compounding the worst smearing penalty, $\times 1.74$, with the worst Hanning factor, $\times 2.36$, gives $\times 4.1$ for those few, a product no released column carries. *For the remaining 1200 windows the quantity cannot be formed at all and the thresholds are unverified on this point:* the ratio of drift ceiling to channel width covers the same range there as in the windows that can be evaluated, but the integration time, 0.6–12 s where it is known, is not in the release.

The primary-beam response, and the two columns that let a reader recompute it

The sensitivity quoted for a window is $S_{\text{min}} = 5\sigma/A$, where A is the primary-beam response at the star’s offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the two differ by A . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades

old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 released windows the offset has a median of $0.089''$, a 99th percentile of $7.43''$ and a maximum of $8.13''$, and A runs from 0.999984 in the median down to 0.5522, below 0.99 on 91 windows and below 0.9 on 34. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$, the first window beyond it 0.35 at $0.62\theta_{\text{PB}}$, and nothing lies between, so the rule binds prospectively and excludes none of the 1651 released windows. Beyond it lie the 12 ϵ Eridani Band 6 windows. 8 of them need a correction of up to $\times 3.5$, past the first sidelobe transition where the Gaussian form fails, and those are withheld: they enter no limit and no row of the released catalogue. The other 4 place the star at the pointing centre and need no correction at all, so there is nothing to withhold them for, and at 3.22 pc the star is too near to leave without a limit. All 4 are in block Xb9c01c_X2004, 0.10 window-hours between them, and none holds a threshold crossing: the largest statistic is 3.8 where the trigger is five. The one fine-channel window, at 488 kHz, reaches 1.8×10^{14} W; the 3 coarse ones at 15.6 MHz reach 9.3×10^{14} – 1.1×10^{15} W. Those four limits are reported here and enter no census statistic. The flux-scale comparison of §4 is a separate continuum measurement on the same star: a block mean over the 4 compact-array executions that place ϵ Eridani at the pointing centre, 0.22 h in Band 6, where no beam correction enters either.

The correction is applied exactly once. The offset is computed for every measurement set from its own FIELD:PHASE_DIR and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only: $S_{\min} = 5\sigma/A$ closes on all 1651 released rows to $8.7\text{e-}05$ in relative terms. The count is in two parts, because on 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\min}$ and flagged as such, so there the closure is arithmetic; the audit is a test on the other 1641 rows, where A comes from the search's own stored geometry and closes to $8.7\text{e-}05$. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The response applied is pb_atten: a Gaussian of full width $1.22\lambda/D$ — an Airy first-null coefficient — at each window's *own* dish diameter, reproduced over all 1651 released rows to 2.6×10^{-5} against 7.7×10^{-2} for the same Gaussian forced to $D = 12$ m. The deposited theta_pb_arcsec is that width at $D = 12$ m for every row, including the 1054 taken on the $1.7\times$ smaller compact-array dishes, so A must be read from pb_atten and cannot be recomputed from it.

The beamwidth convention is a bounded systematic, and the release carries both conventions so that a reader need not take ours. ALMA's primary-beam full width is $1.13\lambda/D$, narrower than the first-null coefficient, so it attenuates more and makes every limit *shallower*: by nothing at all in the median window, by more than one

per cent in 40 of 1651 and by more than ten in 4, the largest being $+10.3$ per cent — those few all LP 736-15 at $8.1''$ off axis. The lowest response in the release moves 0.5522 to 0.5005, still clear of the 0.5 retention floor but by a margin worth saying, and pb_atten_fwhm in the deposit carries that column window by window. Changing the convention changes no disposition, because A divides out of every ratio formed inside a window. The remaining model uncertainty is also bounded by measurement: CASA's blocked-Airy response is better physics than either Gaussian, the largest ratio to it over the whole release is 1.0745, and the counts of systems reaching 10^{15} W are identical under any of the three.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. The displacement decides whether the loss can be treated as a scalar at all: a small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax, and radial velocity where catalogued — leaves each window's own noise unchanged to 0.20 per cent and alters no disposition, but it cannot test the amplitude factor, *because the factor describes a source and those windows hold none.* The statistic at the star is a maximum over a plane of drift trajectories, so moving the extraction point by two thirds of a beam draws a nearly independent noise realisation: the ratio scatters from 0.78 to 1.15 about a median of 0.98, nowhere near the 0.49 the model predicts for a source. A carrier injected at the apparent position and recovered from the *same* visibilities at both positions does return the normalised dirty beam at the omitted displacement, measured on that block's own baselines with no model in it: 0.92 against a predicted 0.89 at 0.25 of a beam, and 0.58 against 0.49 at 0.67. That it exceeds the model at the larger displacement is the search's doing, since maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so a limit corrected by the model is shallower than it needs to be.

The injection campaign is blind to A by construction, and the bias that follows is exactly zero. The injector deposits an unattenuated amplitude at the star's own direction cosines and refers the ladder to the *apparent* 5σ on all 40 scored units, so an injected apparent amplitude recovered nine times in ten implies a true flux larger by $1/A$ while P_{trig} is the power of a true $5\sigma/A$ and A cancels identically; 4 of the units have apparent and corrected $S_{\min}$ differing by more than a part in a thousand, so the cancellation is tested rather than vacuous. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and it reaches only $A \geq 0.9102$, whereas 75 windows on 5 stars lie further off axis, down to $A = 0.552$, so a limit quoted for one of those rests on an extrapolated curve; their 6 crossings are ratios and unaffected.

A few rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconciled: 4 windows of HD 285968 in block A002_Xd98580_X3bf6, whose catalogued σ is 0.88–1.32 per cent *higher* than the surviving product's, so the limits

there are weaker and not stronger. With the 10 windows whose products are not retained at all the declared provenance gap is 14 windows, and the attenuation columns are unaffected.

THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; and were it operating, it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. No mechanism for the displacement is established. What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, the radii being known from the fixed draw, with no free parameter.

Applied out of sample, the correction does not repair the statistic. Frozen — probe radii, bin edges and standardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately. Rescoring the whole survey with the gradient removed moves the list of crossings outranking all their controls from 12 windows to 11, losing 2 and gaining 1, and leaves the survey median rank slightly further from one half than before.

The radial gradient is real, but removing it leaves the median no closer to one half, so the residual is not radial and neither normalisation yields a calibrated tail probability. We therefore keep the frozen rank, read it only as an indication of which crossing to examine first, and rest every disposition on physical evidence or on recurrence. The radial correction, the region-maximum audit and the pseudo-star tests each leave every disposition unchanged, and because the rank gates nothing, none of this enters a sensitivity limit.

One mechanism can be excluded by measurement, and it is the one that would matter most. Equation 1 divides every position by a single noise scale shared by the star and all its probes. A star bright enough to carry multiplicative calibration residuals of its own would then have a noise the shared scale does not describe, and its statistic would be inflated with nothing in the search to reveal it. The ratio of the two noises is measurable from the released control vectors alone: because the star and each probe are divided by the same scale and maximised over the same channel $\times$ drift grid, the stellar statistic divided by the median of the 512 control statistics *is* that ra-

tio, and it is one when the two noises agree. Over the 1595 windows carrying no threshold crossing its median is 1.001 (star-clustered 95 per cent 0.994–1.004), and it does not climb with the brightness of the field, which is what a multiplicative error would scale with: 1.002 in the faintest fifth of control rings against 0.999 in the brightest, and 1.09 over the 8 windows whose ring exceeds 14σ and is therefore set by resolved emission rather than noise. That the ratio *can* rise is shown by the 56 windows that do carry a crossing, where it is 1.18. Part of the reason it does not rise otherwise is structural: a gain error multiplying a continuum raises every channel of an integration by the same factor, and the per-integration median along frequency removes exactly that. **A stellar-position noise denominator would therefore move no statistic in this survey.**

The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block on the pre-registered list deposited with this paper stays in the survey, and no system loses every block — so that the assignment can be recomputed from the UIDs and that list alone. The statistic, the trigger and the recurrence criterion were all fixed before the rule itself, and the first reserved block was searched 3 days later, so every diagnostic built afterwards is incapable of promoting a window. Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Three quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458 over 315 windows, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star in fact outranks all 512 probes in 14 of the survey’s windows against 3.2 if the positions were interchangeable. The direction therefore inflates star-first outcomes rather than suppressing them, which is why the chance expectations of Appendix D are calibrated on the measured rate. Clustering removes neither the shift nor its size: 99 of 149 blocks have a mean rank below 0.5, and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46). The searched sample leans the same way but by less (median 0.49, star-clustered 95 per cent 0.47–0.51), and the two intervals do not overlap, so the displacement is a property of these reserved blocks and is not reproduced in sample. Either way it runs in the direction that inflates star-first outcomes, which is the quantitative reason the rank is read as a diagnostic and never as a significance.

Second, the false-alarm tail factor — the measured rate at which a no-signal position ranks first, divided by the rate exchangeability would give. It is 1.2 on 5 040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. It is a statement about the rank and not about the trigger, so it is reported as a diagnostic; the

TABLE 6

THE BLOCK AND WINDOW LEDGER. EVERY SET OF EXECUTION BLOCKS USED ANYWHERE IN THIS PAPER, WITH THE RESULTS IT ENTERS; THE ROWS SUM, AND EACH INDENTED GROUP SUMS TO THE ROW ABOVE IT. THE PRIMARY CENSUS IS THE SURVEY: UNLESS A STATEMENT NAMES ONE OF THE OTHER SETS, IT IS COMPUTED OVER THAT ROW. HOURS ARE WINDOW-HOURS: ON-SOURCE TIME SUMMED OVER SPECTRAL WINDOWS, SO A BLOCK DELIVERING SEVERAL WINDOWS AT ONCE CONTRIBUTES ITS ON-SOURCE TIME ONCE PER WINDOW. THE CLASSES THEREFORE PARTITION THE HOURS AS THEY PARTITION THE WINDOWS. THE TELESCOPE ON-SOURCE TIME BEHIND THE PRIMARY CENSUS IS 280 h, OF WHICH 152 h CARRIES AT LEAST ONE CLASS A WINDOW AND 276 h AT LEAST ONE CLASS B WINDOW — FIGURES THAT OVERLAP AND MUST NOT BE ADDED. CONFIRMED PLANET COUNTS, HERE AND IN THE DEPOSITED TARGET LIST, ARE TAKEN FROM THE NASA EXOPLANET ARCHIVE **PSCOMP**PARS TABLE (AKESON ET AL. 2013), ACCESSED 2026 OCTOBER 4.

Set of execution blocks	Blocks	Windows	Stars	Systems	Hours	Results it enters
Archival scope: blocks the query exposes	656	—	—	—	—	defines the scope; the sum of the next two rows
... processed with the frozen pipeline	479	—	—	—	—	the second panel below
... not processed at the freeze	177	—	—	—	—	the third panel below
Processed with the frozen pipeline ^c	484	—	—	—	—	the sum of the three rows below
Primary census	404	1651	89	82	1094	every limit, the crossing list, every chance expectation
Class A, fine channels ^a	—	402	65	60	227	the drifting-carrier experiment
Class B, coarse channels ^a	—	1249	88	81	867	the spectral-excess experiment
Reserved hold-out	77	315	36	—	214	the rank displacement, the false-alarm tail factor, the radial control profile
No window survived the quality cut	3	0	0	0	0	nothing
Not processed at the freeze: what became of each	177	—	—	—	—	the sum of the five fates below
Calibrated and searched	144	—	—	—	—	the sum of the four rows below
Archival extension , reported	115	463	16	16	269	recurrence, and the null on data the frozen pipeline had not seen
Searched after the extension was frozen	8	32	—	—	—	nothing; later than the reported extension
Withheld: primary-beam form invalid ^d	8	32	—	—	—	nothing; withheld
No window reached the export	13	0	—	—	—	nothing
No calibration in the ALMA delivery	18	—	—	—	—	nothing; the archival gap
Calibration failed	11	—	—	—	—	nothing; the archival gap
Not attempted	3	—	—	—	—	nothing; the archival gap
Working disk insufficient	1	—	—	—	—	nothing; the archival gap
Beyond 40 pc: out-of-sample control	47	198	22	—	88	the external null; no science count
External calibration sample ^b	326	1322	56	—	—	the rank calibration and the positive control; no science count

^aThe two classes partition the census windows, not its blocks: 252 blocks carry a Class A window and 389 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

^bNot an out-of-sample set, and not an addition to this ledger: every one of its 326 blocks is counted in a row above, 263 in the primary census and 63 in the hold-out, so it is listed apart only to keep it from being counted twice. The harvest it was drawn from holds one further block, toward a star outside the 40 pc work list, and that block is excluded here.

^cThis panel and the scope panel above overlap rather than add: 479 of these 484 blocks are the in-scope row above, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them. 3 of the 5 are in the primary census.

^dEvery one is an ϵ Eri Band 6 block, where the Gaussian primary-beam form is not valid; the same rule withholds 12 windows from the primary census.

quantity the chance expectations need is the exceedance rate at the stellar position, measured on the same blocks in Appendix D.

Third, the radial behaviour of the control ensemble: the standardised level runs $+0.12, +0.10, +0.12, +0.16$ across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

The *external calibration sample* must not be confused with the hold-out, and it is not an out-of-sample set. Its 326 execution blocks were reserved to calibrate the rank and to supply repeats the hold-out does not have, and 263 of them are primary-census blocks and 63 hold-out blocks, with 0 outside both: it enters no count of the science sample because its blocks are already counted there, not because it is external. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls — 4 of them β Pictoris CO — and 3 unattributed against 3.7 expected at the measured tail rate.

One crossing through the whole chain

CP-72 2713 is the clearest worked example, because every step returns a verdict rather than a shrug. In

Band 7 the crossing reaches $T_\star = 5.809$ in channel 35 of its window; it is unattributed, with the one nearby transition and its provenance discussed in Appendix C; both diagnostics favour it, in that it outranks every one of the 512 control positions and the drift-following visibility fit puts the emission at the stellar position, $\text{Re}/\sigma = 5.1027$ against a control-position maximum of 1.8222; and neither of those is a promotion (§5.5).

What disposes of it is the repeat coverage. One of its 5 covering blocks matches the discovery in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two: a carrier radiating at the discovery amplitude through both observations would have returned $T_{\text{pers}} = 6.21$ there, and that block alone puts the pair 4.6σ apart. Combining all 5 gives $T_{\text{pers}} = 6.30$ and 4.8σ , and rules out a carrier above 0.66 of the discovery amplitude. The nearest covering block is 2.05 h away, so this is largely a within-night test: it excludes a carrier of that strength radiating through both epochs and says nothing about an intermittent one. **The disposition rests on the recurrence test and on nothing else**, which is the chain applied to every crossing in the survey.

What the recurrence statistic reads when nothing is there

The exclusion in Table 8 is one formula. Writing S and σ_d for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\text{excl}} = \frac{S - \bar{a}}{\sqrt{\sigma_d^2 + \bar{\sigma}^2}} = \frac{T_{\text{pers}} - T_{\text{rep}}}{\sqrt{1 + (T_{\text{pers}}/T_{\star})^2}}, \quad (\text{B1})$$

with $T_{\text{pers}} = S/\bar{\sigma}$ and $T_{\text{rep}} = \bar{a}/\bar{\sigma}$, so every exclusion follows from three columns of that table and nothing else. T_{rep} is read at the carried drift and at one cell; the largest value over every drift trial at the same cell is tabulated beside it, and because it is a maximum over of order 10^2 trials it is positive on average whatever is there and is never the quantity T_{pers} is compared with. Because the discovery amplitude carries its own error, no exclusion can exceed the strength of the discovery it is built from — and that is also why it is not the strongest statement available. An amplitude selected as the largest of 10^2 to 10^6 cells at $T_{\star} \simeq 5$ is biased high, so a real carrier would be fainter than measured, and the quantity that survives the bias is the limit $(T_{\text{rep}} + 5)/T_{\text{pers}}$: the fraction of the discovery amplitude still ruled out at the trigger.

The same three numbers say where the test has power and where it has none. The criterion asks the repeat to reach the trigger, and the reading there is centred on zero with unit scatter, so a carrier radiating through both observations at the discovery amplitude satisfies it with probability $\Phi(T_{\text{pers}} - 5)$, and at the trigger amplitude $5S/T_{\star}$ — the faintest carrier that could have produced the crossing at all — with $\Phi(5T_{\text{pers}}/T_{\star} - 5)$. Both follow from columns Table 8 already prints, both are tabulated for the two events of Table 4, and both are deposited for every crossing. Where they are small the crossing is untested rather than excluded, and it is reported that way.

The statistic read at the predicted channel is centred on zero and scatters by one, measured and not merely designed. Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is $+0.014$ and its scatter 1.046 ; at the neighbouring channels of the 2 crossings whose repeat spectra survive — random cells, same registration, same drift — $+0.071$ and 0.966 over 638 cells, with 3.19 the largest value anywhere within $\pm 15 \text{ km s}^{-1}$ of their predicted channels; and at the control positions of the discovery windows -0.036 and 1.034 . The largest of the 8 retained control probes of each repeat block averages 1.449 against the 1.424 expected for a maximum of 8, which is the point: **a maximum is positive on average whatever is there.** The largest of the n_{rep} per-block readings averages $+1.102$ over the crossings carrying exclusions, where the null above predicts $+1.130$ for the same n_{rep} , while the combined T_{rep} of Eq. B1 averages -0.119 . Widening the window costs trials and not calibration: the exceedances of §5.6 match what the same accounting predicts, and HD 69830’s windows are the only ones too narrow for the excursion their crossing’s drift implies.

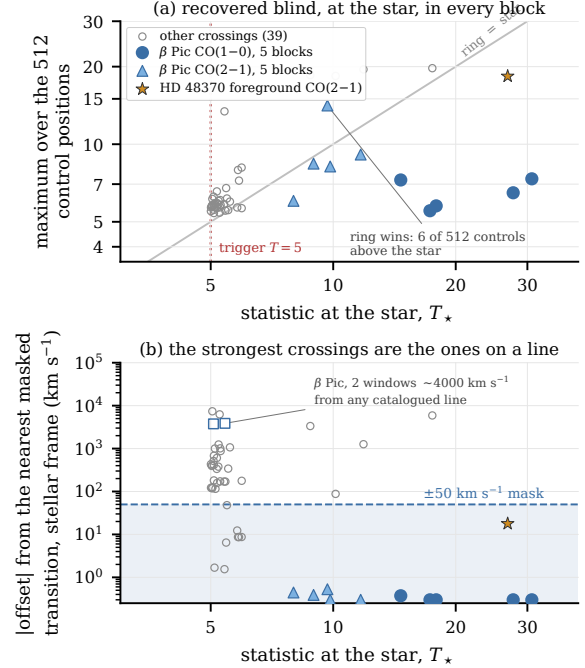


FIG. 10.— The β Pictoris positive control, in both directions. (a) The spatial screen. One point per execution block in which the blind search recovers circumstellar carbon monoxide, plotted as the stellar statistic against the maximum over that window’s 512 control positions: the 9 blocks in which the star outranks every control, and one further block in which the ring wins. *The control is that the search recovers a real narrow line at a stellar position, in two transitions and in every one of these blocks.* It is not that the star outranks the ring everywhere. In block X7116f1_X1785, Band 6, the control ring reaches 14.13 against 9.68 at the star, with 6 of the 512 control positions above it, so the screen rejects a window holding carbon monoxide that is real, centred on the star and independently mapped: **the screen measures compactness and not authenticity** (§5.4). (b) Line confusion, which is what makes the screen’s job hard. Each crossing is plotted by its velocity offset from the nearest masked transition *in its own star’s rest frame*, with the adopted $\pm 50 \text{ km s}^{-1}$ half-width marked. The 10 β Pictoris carbon-monoxide crossings land $0.1\text{--}0.5 \text{ km s}^{-1}$ from CO, at the resolution of the systemic velocity itself, while the two CS(5 $\rightarrow$ 4) crossings of the same star sit $3765\text{--}3870 \text{ km s}^{-1}$ from anything catalogued and are not attributed at any width. 39 further crossings are drawn in grey, with HD 48370 marked; 52 of the 56 are plotted, the remainder being crossings for which the release stores no control ring. *A crossing is identified by where it falls in this plane and not by how strong it is*, which is why the line mask and not the trigger is the productive test.

THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. The labelled crossings are tested for recurrence below along with the rest.

The list, and when it was made. The mask holds every transition of the 12 species these spectral setups cover — the three CO isotopologues, HCN, HCO^+ , CS, CN, SiO, H_2CO , SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table 7). No strength cut is applied across species: the Einstein coefficients of the CO ladder are $10^{-7}\text{--}10^{-6} \text{ s}^{-1}$, so any threshold admitting the strong lines of HCN would discard the brightest lines in these fields. Within a species the coefficient keeps a rotational transition with its main

TABLE 7

THE LINE MASK, BY SPECIES: THE NUMBER OF TRANSITIONS N IT HOLDS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES. HYDROGEN RECOMBINATION LINES CARRY NO TABULATED E_u . THE TRANSITION-BY-TRANSITION LIST IS IN THE DATA RELEASE.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03–231.90	
SO	15	100.03–682.68	16–143
H ₂ CO	13	101.33–491.97	10–106
C ¹⁸ O	3	109.78–329.33	5–32
CN	50	113.12–680.30	5–114
CO	3	115.27–345.80	6–33
CS	3	195.95–342.88	24–66
SiO	2	217.10–347.33	31–75
¹³ CO	8	220.40–330.59	16–32
HCN	7	354.50–354.51	43–43
HCO ⁺	1	356.73–356.73	43
[C I]	1	492.16–492.16	24

hyperfine components and drops the weak satellites, and the rarer isotopologues are not named at all and would add 525 transitions. The list is a single Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on 2026 October 6 and frozen with the data release, so it is reproducible as a file rather than as a description. It post-dates the 2026 September 11 fixing of the disposition criteria, and so does every species in it; no species is privileged by having been thought of first, and the dispositions turn instead on the physics of each coincidence, set out beside it.

The frame is the star’s own, and the sign is not the usual one. A crossing’s frequency is carried from the topocentric value to the barycentre using its own block’s pointing and mid-time, and then to the stellar rest frame using the star’s systemic velocity. The barycentric term runs from -29.2 to $+29.2$ km s^{−1} across the survey, so a tolerance applied in the observed frame would spend 58 km s^{−1} of its budget on the Earth’s motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: 12 km s^{−1} apart as observed, they agree to 0.03 km s^{−1} in the star’s frame. Throughout, $\Delta v_\star = c(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency — the opposite sign to the velocity convention of a spectral line.

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: it follows from Keplerian motion where these molecules survive, $\lesssim 13$ km s^{−1} beyond 10 au, and is ± 50 km s^{−1}. At that width the mask removes 5.04 GHz of the 118.1 GHz unique-frequency union, 4.3 per cent of it, concentrated in the line-tuned fine-channel windows; the cost band by band is in the data release. Between ± 20 and ± 100 km s^{−1} the mask labels 17 to 19 of the 56 crossings, and the 2 unattributed crossings that outrank all 512 control positions in their own windows are the same two at every width.

Sulphur monoxide is reported and not acted on. CD−57 1054, CP−72 2713 and HD 161868 carry crossings -1.7 , -12.3 and -48.0 km s^{−1} from sulphur monoxide transitions of upper-state energy 21–87 K, each inside a Band 7 window that also covers CO(3 → 2), and none has a nearer transition of any other masked species. Two things argue against reading those coincidences as emission. Sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O; and toward CP−72 2713 the CO(3 → 2) line falls in the

same window and is absent to 10.5 mJy in a 0.42 km s^{−1} channel, while the crossing itself is 12.2 mJy — a sulphur monoxide line outshining an undetected CO(3 → 2) line in the same data is not a plausible object. Both arguments are about the species and not about any one crossing, so all 3 are treated alike: the offset is printed beside each in Table 8 and disposes of none of them, giving 18 attributed crossings against 38 unattributed, of which two outrank every control position in their own windows. Reading the coincidences as attributions instead would give 21 and 35, and 1 such crossing. The recurrence test gives the same answer for all three either way — 4.1–5.5 σ against a carrier that stayed inside the repeat window and radiated through both epochs — so the choice moves counts and no conclusion.

The labelled crossings are tested for recurrence too. Of the 18 attributed crossings, 15 have a later execution block covering the predicted cell, and 13 are present again: toward β Pic and HD 285968 a later block carries its own threshold crossing at the same stellar-frame frequency, at $T_\star = 5.9$ to 30.8. That is the test working in the positive direction on emission that is certainly real, and no exclusion is formed for those rows, because what it would exclude has just been observed. The other 2 are not recovered, and for them the repeat windows bound the statistic at the predicted cell from above and the exclusion from below; both are printed as bounds. A resolved line need not reappear at the same brightness in a block of different baseline coverage, so those two bounds bear on the attribution rather than on the presence of a carrier. The remaining 3 cannot be tested and are named rather than counted: HD 139084 B, HD 48370 and HIP 10679, each observed once at the frequency concerned.

Image-sideband leakage does not account for any crossing. The mask holds signal-sideband frequencies, so a bright line in the image sideband could in principle be mistaken for a carrier. The question arises for 2 unattributed crossings, at 241.75 and 241.84 GHz toward β Pic, each sharing a block with a CO(2 → 1) crossing at 230.52 GHz in the opposite sideband. A mirror about the first local oscillator would place that line on the crossing only for an oscillator at 236.13–236.18 GHz, where the block’s own spectral windows require 235.02–235.68 GHz: at the setting the mirror needs, the block’s lowest window would sit at an intermediate frequency of 10.5 GHz, beyond the 10.0 GHz the receiver provides (Cortes et al. 2024). Their image frequencies fall at least 1353 km s^{−1} from every masked transition.

A wider list is not a better one. Every Splatalogue transition over the searched intervals under the same energy cut and with an Einstein coefficient above 10^{−5} s^{−1} — 5,114 transitions of 235 species — masks 75.1 per cent of the union and moves 27 of the 56 crossings across the line, one of them a crossing that outranks every control, at -36 km s^{−1} from a sulphur dioxide transition. At these frequencies almost any frequency lies within ± 50 km s^{−1} of something in such a list, which is why the species are restricted to what these spectral setups cover, and why the outcome is a label rather than a verdict.

THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

The paper’s central claim is that the unattributed crossings are no more numerous than chance allows. This appendix gives that expectation, what it is conditioned on, and what the control ensemble calibrating it can support.

What the expectation is conditioned on

The expectation is built from the control positions, the only empirical null the data provide, and it imposes the condition an event must actually satisfy: to reach the trigger, and to fall outside the attribution windows. Each Class A window contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so — no cell count enters it and no form is assumed for the tail — and summed over the windows 37.5 are expected to trigger by chance. The share of its own channel grid lying inside an attribution window, in that star’s rest frame, is the probability that such a position would be attributed, a control maximum being free to fall on any channel. Weighted by the first that share is 20 per cent, against a median of 6.0 per cent per window, because 15 narrow line-tuned windows lie wholly inside the mask and their fields, which hold real emission, are where a control position most often triggers. What is left is 29.9 expected unattributed Class A crossings against 34 observed, under the same condition and over the same windows, reduced by the mask that classifies them.

Both sides run over the 417 Class A windows the survey covers. The released catalogue tabulates 402 of them window by window; the remaining 15 belong to the archival tail, are carried at the mean rate of the rest, and hold the two unattributed crossings toward HD 14055. A systemic velocity is measured for the stars of 277 of the tabulated windows and the others are taken at rest, which moves the masked share by 3.0 points. No window holds more than one crossing, so both sides count windows.

The one assumption that cannot be read off the searched data is that a star behaves like a control position in its own field, and the reserved blocks measure that in the quantity used here. Their stars reach the trigger in 8 of 62 fine-channel windows where the control positions of those same windows predict 7.5: a ratio of 1.07, with 0.48–1.79 allowed by resampling the reserved blocks, and no two of the events in one block. In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. No departure is detected and none is applied, and the limit is the one so few events allow: a departure smaller than about 35 per cent would not have been seen, and that is the accuracy of every chance expectation in this paper.

The 13–50 quoted beside the expectation is what 20 000 resamplings of the execution blocks allow, each window’s rate redrawn from its own 512 probes: windows share blocks and are not independent trials, and the ensemble is finite. Those two terms alone give 27–33, so what widens the range is the calibration above — which is why the comparison carries no limit on an added population.

Every Class B crossing in the survey comes from the

one execution block of §D.2. Setting that block aside the coarse class adds almost nothing to either side — the expectation over both classes together is 30.1 against the same 34 observed — so the Class A comparison is also the survey’s: only 455 of the 1651 windows hold any control position above $T_{\text{trig}} = 5$ at all, and a coarse single-pointing window has too few cells for its own field to reach the trigger by chance.

The statistic is always the one measured over the window’s whole field. 157 windows admit integrations lying inside the primary beam that a narrower field would discard; keeping them recovers a median 2.1 times the on-source time and moves the search statistic by a median 0.07. The whole-field value is the one adopted, on 8 rows of Table 8, and a window’s control ensemble is measured with it, so neither side of the comparison above mixes two fields. Where the deposited catalogue’s own disposition column disagrees — on 4 crossings that fall below the trigger, and the species list moves none into attribution — the ledger is the authority.

Execution-block quality

The comparison with the control positions assumes each block’s ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition is therefore applied to every block rather than to the one that prompted it.

Two criteria are built from that statistic and they are not the same test. The statistic is the median of a window’s own 512-position control ensemble, which separates a field with a source in it from a field that is no null at all. A *window* is flagged when that median exceeds twice the median of the same quantity over the survey, 5.5 against 2.74; the flag is a property of the window and cuts across attribution by design, firing on 5 of the 1651 windows in 2 blocks toward 2 stars (BD+05 1668 and HD 48370), and every one of those windows carries a crossing, attributed and unattributed alike. A *block* is judged on the same median divided by the median over the windows of its own class, since a maximum over a larger channel×drift grid sits higher and one threshold should mean the same thing in both; a block fails when the median of that ratio over its windows exceeds twice the scale of its class. The factor is the same in both and both were fixed before either was applied to anything.

Applied to all 404 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 against 1.12 for the next worst. Nothing else is close, and the unscaled form of the rule fails the same 1 block. Its 4 windows are all coarse-channel and hold the survey’s entire Class B crossing population, so it carries 4 of the 56 threshold crossings and no Class A statistic here has ever contained it. The population the paper characterises is therefore the 52 crossings in quality-passing data: 18 attributed and 34 unattributed, 2 of those outranking all 512 control positions of their own window against 0 in the failed block. The other 4 appear in Table 8 with their block named and enter no count elsewhere.

The control ensemble: its calibration, and its resolution

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Over the 1 645 census

windows that retain a control vector the stellar ranks are consistent with the one half exchangeability implies — median 0.48 with each star carrying equal weight, 0.47 to 0.51 — in both array strata, the compact-array windows measured on each observation’s own antenna table rather than on a 12m beam. That is a weak statement and not a strong one, since a shift smaller than about 0.042 could not have been seen in a sample of this size. The reserved blocks lean further from one half; Appendix B.1 compares the two samples, and the expectations above do not rest on their agreeing, being summed from the measured exceedance rate at the stellar position itself.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions of the same field decides which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. A survey of 1651 windows would need a rank 64 times finer, and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus carries only $N_{\text{eff}} \simeq 30\text{--}43$ independent positions and resolves no better than 0.023–0.032, with the primary beam bounding how many a single pointing contains. That floor is the design’s central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence.

The trials budget

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise, their maxima sitting at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian — a budget predicting 476 windows with a control above T_{trig} against the 455 observed — while fitting each window’s effective count against its own ensemble returns 0.81 of its searched cells. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.5 rather than 37.5, 3 per cent away.

The on-star crossings do exceed that budget, and the attribution accounts for it. A Gaussian tail over every searched cell predicts 37.8 stellar exceedances against the 56 found, but 18 are identified with a catalogued transition and 4 come from the block of §D.2, leaving 34 against the 30.1 predicted. No excess survives the attribution.

One stratum departs from uniformity and it is the one that matters. Counting the windows whose stellar peak both reaches the trigger and exceeds all 512 control positions of its own window, the 402 fine-channel windows hold 12 against 0.78 expected, while the coarse stratum, three times larger and predicting 2.43, contains 0. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and an unresolved carrier would appear in Class A first for the same reason. The departure survives removal of every fine window carrying a stellar crossing, so fine-window rank probabilities are optimistic for the class as a whole. Without the trigger condition the count is a dif-

ferent one: 2 further coarse windows have a star above every probe at 4.20 and 4.36, bringing it to 14 against 3.2 over all 1651 windows. One quantity, two conditions; the chance comparison uses the conditioned one.

What is left is bounded, and the bounds are null. The real stellar ranks agree at $p = 0.38$ with the 220 672 ranks each control takes against the other probes of its own window; radial suppression is below 10 per cent in the mean; and scrambling a window so nothing coherent survives along a drift track carries 4 past the rank floor and changes no disposition.

RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has, and the coverage it does not have, are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3; 1 of the 1651 searched windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths, that the array sits at 5000 m on a site chosen partly for radio quietness, and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre does not fringe-track but acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

The coverage of the screening is uneven, and that is stated rather than smoothed over. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window from the released catalogue, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correlator flagging is the observatory’s own, replayed from each block’s delivered calibration script, but it is not recorded per window in the release. *No per-channel interference flag of our own exists, and none can be recovered.* Three measurements follow, and each is a test the population could have failed.

First, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster — but on the survey’s own molecular lines. The test needs a measured peak frequency in the frame in which an interfering carrier would stand still, so it runs topocentrically over the 51 of the 56 crossings for which the release records one, against the transitions the search was tuned to. For the other 5, all β Pic carbon-monoxide crossings, the release gives the velocity offset from the line instead, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of

TABLE 8

THE CROSSING LEDGER, FROM WHICH EVERY DISPOSITION IN THIS PAPER CAN BE REPRODUCED. ALL 56 THRESHOLD CROSSINGS, INCLUDING THE 4 IN THE FAILING BLOCK OF §5.3.2: THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT AND THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC T_* , THE LARGEST STATISTIC T_{ctrl} AMONG THE WINDOW'S 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION AND THE STELLAR-FRAME OFFSET Δv_* FROM IT, THEN THE RECURRENCE TEST: n_{rep} COVERING REPEAT BLOCKS SPANNING Δt DAYS, THE STATISTIC T_{pers} A CARRIER PERSISTING AT THE DISCOVERY FLUX WOULD HAVE PRODUCED AT THE PREDICTED STELLAR-FRAME CELL, THE STATISTIC T_{rep} MEASURED THERE AT THE CARRIED DRIFT, AND THE EXCLUSION OF EQ. B1, WHICH THREE OF THESE COLUMNS REPRODUCE, TWO OF THEM ALSO GIVING THE LIMIT DEFINED IN APPENDIX B. T_* AND T_{ctrl} COME FROM THE SAME MEASUREMENT OF THE SAME WINDOW; T_{ctrl} GATES NOTHING AND SAYS ONLY WHICH CROSSING WAS EXAMINED FIRST, AND THE DRIFT-MAXIMISED READING OF THE CELL IS IN TABLE 4 FOR THE TWO LEADING EVENTS AND IN THE DATA RELEASE FOR THE REST. ROWS WHOSE WINDOW IS DATA-QUALITY FLAGGED (§D.2) ARE MARKED, AND ON THOSE T_{pers} AND THE EXCLUSION ARE WITHHELD: BOTH ARE BUILT FROM THE DISCOVERY AMPLITUDE, AND IN A WINDOW WHOSE CONTROL RING STANDS AS HIGH AS ITS STAR THAT AMPLITUDE CANNOT BE ASSIGNED TO A POINT SOURCE AT THE STAR. 1 OF THOSE ROWS LIES OUTSIDE THE FAILING BLOCK — HD 48370, A RESOLVED DISC — AND IS WITHHELD FOR THAT REASON AND NOT FOR ITS BLOCK. A CROSSING IS ATTRIBUTED WHERE $|\Delta v_*| \leq 50 \text{ km s}^{-1}$; THE TRANSITION NAMED IS THE NEAREST IN THAT FRAME, SO FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. BLOCK IDENTIFIERS OMIT THE CONSTANT A002_ PREFIX.

Star	block	$B \nu_{\text{topo}}$	ν_* (GHz)	T_* T_{ctrl} line	Δv_* n_{rep}	Δt (d)	T_{pers}	T_{rep} σ_{excl}	disposition	
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.105473.39874.69 CS(5-4)	-5911.9	29	0.9-113	n/a	+0.15	n/a unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.745848.30383.74 H ₂ CO(3 _{1,2} -2 _{1,1})	-1264.5	29	0.9-113	n/a	+0.37	n/a unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.745841.35575.48 CN(2 _{3/2} -1 _{1/2})	+88.0	29	0.9-113	n/a	+0.66	n/a unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.199239.58683.29 CS(5-4)	-3349.2	29	0.9-113	n/a	+0.77	n/a unattributed; flagged
β Pic	Xf5d76d_Xeb8	3	115.2603	115.271130.765 7.34 CO(1-0)	-0.2	4	0.0-53	n/a	+27.68	n/a attributed; recurs
β Pic	Xf5d76d_Xe19	3	115.2603	115.271127.684 6.48 CO(1-0)	-0.2	4	0.0-53	n/a	+30.76	n/a attributed; recurs
HD 48370	Xc26103_X155a	6	230.5117	230.524326.834 18.36 CO(2-1)	-17.8	—	—	n/a	—	n/a attributed; flagged
β Pic	Xf4df6f_Xc7	3	115.2620	115.271117.907 5.76 CO(1-0)	-0.2	4	53	n/a	+30.76	n/a attributed; recurs
β Pic	Xf5d76d_Xcc5	3	115.2604	115.271117.288 5.52 CO(1-0)	-0.2	4	0.0-53	n/a	+30.76	n/a attributed; recurs
β Pic	Xf5d76d_X32f1	3	115.2606	115.271314.654 7.27 CO(1-0)	+0.4	4	0.0-53	n/a	+30.76	n/a attributed; recurs
β Pic	Xd9668b_X3a90	6	230.5161	230.537811.681 9.12 CO(2-1)	-0.3	5	0.0-1983	n/a	+9.83	n/a attributed; recurs
β Pic	Xf61341_X1484	6	230.5284	230.5379 9.832 8.20 CO(2-1)	-0.1	5	25-1983	n/a	+11.68	n/a attributed; recurs
β Pic	X7116f1_X1785	6	230.5272	230.5384 9.677 14.13 CO(2-1)	+0.5	5	25-1983	n/a	+11.68	n/a attributed; recurs
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377 8.948 8.41 CO(2-1)	-0.4	5	0.0-1983	n/a	+11.68	n/a attributed; recurs
β Pic	Xa7a216_X23e4	6	230.5273	230.5377 7.988 6.03 CO(2-1)	-0.4	5	5601-1312	n/a	+11.68	n/a attributed; recurs
61 Vir	Xc079b5_X82f	7	346.3138	346.3232 5.958 5.65 SO(9 ₈ -7 ₇)	-177.6	3	0.1-3.0	9.7	+0.39	4.9 unattributed; no recurrence
HD 285968	Xd98580_X3bf6	6	230.5021	230.5447 5.956 8.12 CO(2-1)	+8.7	3	0.0	n/a	+5.87	n/a attributed; recurs
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447 5.868 7.22 CO(2-1)	+8.7	3	0.0	n/a	+5.96	n/a attributed; recurs
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447 5.835 7.94 CO(2-1)	+8.7	3	0.0	n/a	+5.96	n/a attributed; recurs
CP-72 2713	Xf0235_X4a6d	7	344.2697	344.2965 5.809 5.68 SO(8 ₈ -7 ₇)	-12.3	5	0.1-2.1	6.3	-0.82	4.8 unattributed; no recurrence
HD 139084	B Xcd8029_Xb6b0	7	345.8241	345.8241 5.763 17.70 CO(3-2)	+24.4	—	—	—	—	—
HD 14055	Xff3d8_Xd46	7	331.7699	331.7664 5.575 5.99 ¹³ CO(3-2)	+1068.7	30	0.9-650	170.4	-0.02	5.6 unattributed; no recurrence
HD 23484	X10b22d7_X2445	6	230.7904	230.8006 5.520 5.55 CO(2-1)	+341.5	4	0.1-2.1	10.4	+1.25	4.3 unattributed; no recurrence
HD 161868	Xff45a6_X10ca8	7	345.6338	345.6491 5.481 6.16 SO(2 ₃ -2 ₁)	-48.0	13	0.0-6.1	92.2	+0.24	5.5 unattributed; no recurrence
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035 5.456 6.76 ¹³ CO(2-1)	+6.5	—	—	—	—	—
HD 14055	X11adad7_X25c5e7	7	345.4564	345.4303 5.448 6.82 SO(2 ₃ -2 ₁)	-237.8	30	0.1-653	14.0	+0.03	5.1 unattributed; no recurrence
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070 5.433 5.52 CO(2-1)	-170.3	4	0.8-425	69.7	-0.93	5.5 unattributed; no recurrence
β Pic	Xd9668b_X3a90	6	241.7512	241.7739 5.420 5.81 CS(5-4)	-3869.7	1	1.0	5.7	+0.72	3.4 unattributed; no recurrence

TABLE 9

The crossing ledger, continued. THE REMAINING 28 OF THE 56 THRESHOLD CROSSINGS, IN THE SAME ORDER AND WITH THE SAME COLUMNS AS TABLE 8, WHOSE CAPTION DEFINES THEM.

Star	block	$B \nu_{\text{topo}}$ (GHz)	ν_* (GHz)	T_* T_{ctrl} line	$\Delta \nu_*$ n_{rep}	Δt (d)	T_{pers}	T_{rep} σ_{excl}	disposition	
HR 1010	Xc6fa08_X1053	6	230.5133	230.52875.410 5.65 CO(2-1)	-12.0	7	13-14	60.1	<+4.60	>5.0 attributed; not recovered
ALMA J153702653-33192492	Xf0235_X8392	6	230.5222	230.53925.405 13.41 CO(2-1)	+1.6	6	0.0-3.8	6.2	<+4.15	>1.4 attributed; not recovered
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.43575.393 5.85 C ¹⁸ O(2-1)	-170.2	1	10	6.7	+0.62	3.8 unattributed; no recurrence
HD 131392	X129754b_X6946	6	231.1998	231.22315.289 5.81 H(30a)	-876.3	8	0.1-41	15.7	-0.26	5.1 unattributed; no recurrence
HD 14055	Xf99e1_X24be0	7	331.7078	331.70355.277 5.69 ¹³ CO(3-2)	+1011.6	30	0.9-652	252.2	+0.15	5.3 unattributed; no recurrence
AU Mic	Xa42f75_X319	6	230.6864	230.67055.274 6.07 CO(2-1)	+172.3	2310-455	6.4	-0.16	4.2 unattributed; no recurrence	
HD 69830	X12209f7_X16edb7	7	322.4325	322.44445.259 5.42 C ¹⁸ O(3-2)	-6268.5	5	0.1-29	8.9	-0.48	4.8 unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	216.1844	216.20365.228 6.02 SiO(5-4)	-1244.6	7	18-126	11.0	+1.03	4.3 unattributed; no recurrence
AU Mic	X899169_X1838	6	230.8252	230.82945.221 5.85 CO(2-1)	+379.0	2145-310	10.0	+0.19	4.5 unattributed; no recurrence	
HD 14055	Xf99e1_X2c3bb7	7	345.5256	345.52165.186 5.66 SO(2 ₃ -2 ₁)	-158.6	30	0.1-651	172.8	+2.00	5.1 unattributed; no recurrence
HD 14055	Xffde1_X3835	7	345.0099	345.00705.177 5.66 SO(2 ₃ -2 ₁)	-604.8	30	0.1-649	187.2	-1.78	5.2 unattributed; no recurrence
HD 14055	Xf99e1_X2b2ee	7	344.7007	344.69655.175 5.92 SO(8 ₈ -7 ₇)	+336.0	30	0.1-651	210.7	-0.00	5.2 unattributed; no recurrence
LHS 1140	Xd9d7c7f_X6fd2	6	230.6355	230.62685.136 5.68 CO(2-1)	+115.5	4	5.0-27	7.3	+0.33	4.0 unattributed; no recurrence
HD 10647	Xf294f_Xaf70	7	331.6570	331.69435.120 5.91 ¹³ CO(3-2)	+1003.3	10	0.1-592	125.8	+0.39	5.1 unattributed; no recurrence
CD-57 1054	Xcf525f_X5aa4	7	346.5099	346.52655.108 5.71 SO(9 ₈ -8 ₇)	-1.7	1	39	6.0	-0.27	4.1 unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	217.72695.091 5.41 H ₂ CO(3 _{0,3} -2 _{0,2})	-680.5	7	3.8-242	11.6	+0.07	4.6 unattributed; no recurrence
GJ 849	Xd9668b_X841a	6	230.8769	230.85815.090 5.83 CO(2-1)	+416.3	5	0.1-51	9.3	-1.69	5.3 unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	218.8729	218.89235.090 5.56 H ₂ CO(3 _{2,1} -2 _{2,0})	+181.2	7	18-126	10.5	-1.04	5.0 unattributed; no recurrence
HD 14055	X1001fdc_X29ae	7	331.1559	331.15465.082 6.26 ¹³ CO(3-2)	+513.8	—	—	—	—	—
β Pic	Xd9668b_Xa9df	6	241.8367	241.85945.079 5.99 CS(5-4)	-3765.1	1	1.0	4.8	-0.63	3.9 unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	218.66885.048 5.77 H ₂ CO(3 _{2,1} -2 _{2,0})	-125.1	7	3.8-242	11.6	+0.00	4.6 unattributed; no recurrence
HD 69830	X12277ed_X178c6	7	321.2005	321.21495.046 6.13 C ¹⁸ O(3-2)	-7387.7	5	6.0-36	10.2	-0.46	4.7 unattributed; no recurrence
HR 1010	Xc66ce1_Xaa9	6	230.8348	230.84915.044 5.72 CO(2-1)	+404.6	7	1.1-14	10.3	-1.53	5.2 unattributed; no recurrence
61 Vir	Xc067f7_X822	7	345.5599	345.56785.038 5.79 SO(2 ₃ -2 ₁)	-118.5	3	2.0-3.0	9.3	+0.45	4.2 unattributed; no recurrence
HD 170773	Xf0235_X8c42	7	331.0101	331.02345.029 5.75 ¹³ CO(3-2)	+394.8	4	0.9-113	74.8	+1.56	4.9 unattributed; no recurrence
ALMA J153702653-33192492	Xd9d7c7f_Xbb64	7	347.2179	347.18905.009 5.68 SiO(8-7)	-122.2	1	0.1	5.0	-2.06	5.0 unattributed; no recurrence
HD 14055	X1190de8_Xd526	7	344.4676	344.45155.006 5.96 SO(8 ₈ -7 ₇)	+122.7	30	20-611	11.6	-0.60	4.8 unattributed; no recurrence
η Crv	X122b6ff_X1041e	7	357.2815	357.24755.006 5.50 HCO ⁺ (4-3)	+431.3	1	87	11.0	-0.62	4.8 unattributed; no recurrence

the 51, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected if each window's peak fell anywhere on that window's own channel grid, $p = 0.0007$. The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2 $\rightarrow$ 1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ($p = 0.111$). *The concentration is the attribution, not a residue left over after it.* Table 8 counts the same band as 14 crossings with 11

attributed, a different test over a different set — all 56, each attributed in its own star's rest frame — and the 2 it adds are the ones whose peak frequency this test has no measurement of.

Second, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1397 of the 1651,

of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks' fractional positions within their own windows have median 0.50 and are uniform.

Third, a cross-target occupancy screen over every searched window and not only the crossings. Interference belongs to the observatory and not to the star, so a terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. It does not even for the leading event: the one unattributed rank-passing event has a released crossing frequency; 40 further windows, toward 24 other stars in 32 other blocks, cover that frequency, and none of them carries a crossing in the same channel. Over all 3027 searched windows toward 133 stars, with each window's peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9 , i.e. *below* the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3 . There is no excess to explain. Two features of the screen matter. It is keyed on the star and not on the product directory, because a star's own repeat observations are not different targets; and 35 of those 109 windows sit on a masked transition (CO(2–1) $\times 17$, CO(1–0) $\times 10$, CO(3–2) $\times 4$, $^{13}\text{CO}(2-1) \times 3$, SiO(5–4) $\times 1$), so the mask has to be applied first — a carbon monoxide line falls at nearly the same sky frequency toward every nearby star, which is exactly the coincidence the screen is looking for.

One execution block fails these tests, and it fails them as a block rather than as a carrier. Its 4 crossings toward BD+05 1668 are the highest statistics in the survey outside β Pictoris and HD 48370, and the field statistics that identify the failure are in §5.3.2. What belongs here is whether interference accounts for it, and three readings say not. The visibility fit returns a large imaginary part, placing the emission away from the phase centre. The block's 24 sibling executions of the same field, with the same array and the same spectral setup, produce no crossing at all. And it is not a carrier at a fixed sky frequency: each of the four frequencies is covered by 127–

137 further windows toward 17–22 other stars, and 0 of them carries a crossing in the same channel.

One reading does point at a mechanism, and it can be tested across the survey. 2 of the 4 sit at the identical channel of two different spectral windows, which is the signature of a fixed intermediate frequency — and a feature with no geometric fringe phase, a spur common to every antenna being the obvious case, does not fringe-track, so it gathers at the phase centre rather than at the star. Here the two are a twentieth of an arcsecond apart, so the block cannot test its own hypothesis; the survey can, because the star lies 5.0–8.1 arcsec from the phase centre in 75 windows whose control annulus brackets it, putting a median of 15 probes within two synthesised beams of the phase centre and the star well outside. Against the probes at the same distance from the star but elsewhere in azimuth — matching the radius matters, or a centrally concentrated disc would imitate the effect — those probes are higher by $+0.09 \pm 0.08$ in the statistic, consistent with zero and bounded by 0.26, smaller by a factor of about 33 than the 8.4 by which this block's ring is raised. *The phase centre is not where the excess lives.* The evidence supports a data-quality failure of one execution block, and that is how these 4 crossings are dispositioned (§5.3.2). Two criteria are involved and they are not the same one (Appendix D.2): the window-level flag, which fires on 5 of the 1651 windows toward 2 stars (BD+05 1668 and HD 48370), the second of them a resolved disc rather than a defect; and the block criterion built from it, which fails 1 block of the 404 searched. It is the block criterion that disposes of these crossings.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. We claim only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. The three tests establish that the crossing population as a whole carries none of the signatures interference leaves, which is a statement about the population and not a clearance of any individual crossing.

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